Article in Press

https://doi.org/10.1038/s44387-026-00166-3

Financial time-series generation: a systematic survey of taxonomy, evaluation, datasets, and open challenges

Check for updates

Yingxiao Zhang[1,2], Jiaxin Duan[3], Yue Zou[1,2], Ke Feng[1] ✉ & Jing Xiao[4] ✉

The rapid advancement of generative artificial intelligence has catalyzed a paradigm shift in quantitative finance, transitioning from traditional point forecasting toward comprehensive financial time-series generation (FTSG) across extrapolation, imputation, and full-scale data synthesis. This transition is motivated by the critical need to simulate rare market regimes, reconstruct incomplete historical records, and produce privacy-preserving synthetic environments for stress testing and trading execution. However, empirical financial properties—including heavy tails, volatility clustering, non-stationarity, and regime shifts—pose severe hurdles for generic generative frameworks. In this systematic survey, we review 131 studies from 2021 to early 2026 to bridge the gap between machine learning capabilities and econometric validity. We introduce a two-level taxonomy structuring the domain across task formulations and algorithmic families, and establish a multi-dimensional evaluation protocol encompassing distributional fidelity, stylized facts preservation, and downstream utility. Furthermore, we catalog curated financial databases and benchmark suites while addressing empirical dataset construction pitfalls. Finally, we identify pivotal open challenges in causal representation learning, econometrically bounded modeling, and online adaptation under distribution shifts. This review provides a standardized roadmap for developing robust, domain-aligned financial generative models.

Time-series analysis (TSA)[1, 2] has long been a central pillar of quantitative finance[3, 4], providing the mathematical foundation for modeling market dynamics, assessing risk, and pricing assets. Traditionally, TSA in finance was closely associated with forecasting[5]: predicting the next price movement, the future volatility level, or the impending macroeconomic shift. Modern probabilistic forecasting goes beyond point estimates by quantifying predictive distributions and forecast uncertainty[6]. As the financial industry becomes increasingly data-driven and algorithmic, the demand has evolved from point estimation to the comprehensive generation of plausible market scenarios. This paradigm shift has given rise to Financial Time-Series Generation (FTSG), which extends conventional forecasting to include fully synthetic financial sequences[7–9], missing-data imputation[10–12], and probabilistic extrapolation of future financial trajectories[13].

The motivation for FTSG is deeply rooted in the structural limitations of real-world financial data. Unlike text or images, financial markets are characterized by a low signal-to-noise ratio, extreme non-stationarity, and abrupt regime shifts triggered by geopolitical events or macroeconomic policy changes. Furthermore, empirical financial distributions frequently exhibit "stylized facts" such as heavy tails, volatility clustering, and leverage effects[14]. While historical data contains instances of these phenomena, critical events like market crashes or sudden liquidity droughts are inherently rare. Consequently, discriminative models trained solely on historical records often suffer from overfitting and fail to generalize during crises. FTSG can address this by creating high-fidelity-and, when trained with formal privacy controls, potentially privacy-preserving-synthetic data that can augment training sets, simulate counterfactual stress-testing scenarios, and train robust reinforcement learning agents in simulated environments.

Across the task taxonomy of this survey, we use FTSG to denote the umbrella research domain (Financial Time-Series Generation), while FTSE, FTSI, and FTSS denote the three task-level formulations for extrapolation, imputation, and synthesis, respectively. This terminology is used consistently throughout the paper and is summarized again in the Problem Formulation and Taxonomy and Applications sections.

Generative AI has rapidly entered financial modeling, as illustrated in Fig. 1. The main families include generative adversarial networks (GANs)[7–9, 15], denoising diffusion probabilistic models (DDPMs)[13, 16],

[1]School of Software and Microelectronics, Peking University, Beijing, China. [2]Pingan Bank Co., Ltd., Shenzhen, China. [3]China Electronics Cloud Technology Co., Ltd., Beijing, China. [4]Pingan Group Co., Ltd., Shenzhen, China. ✉e-mail: fengke@pku.edu.cn; xiaojing661@pingan.com.cn

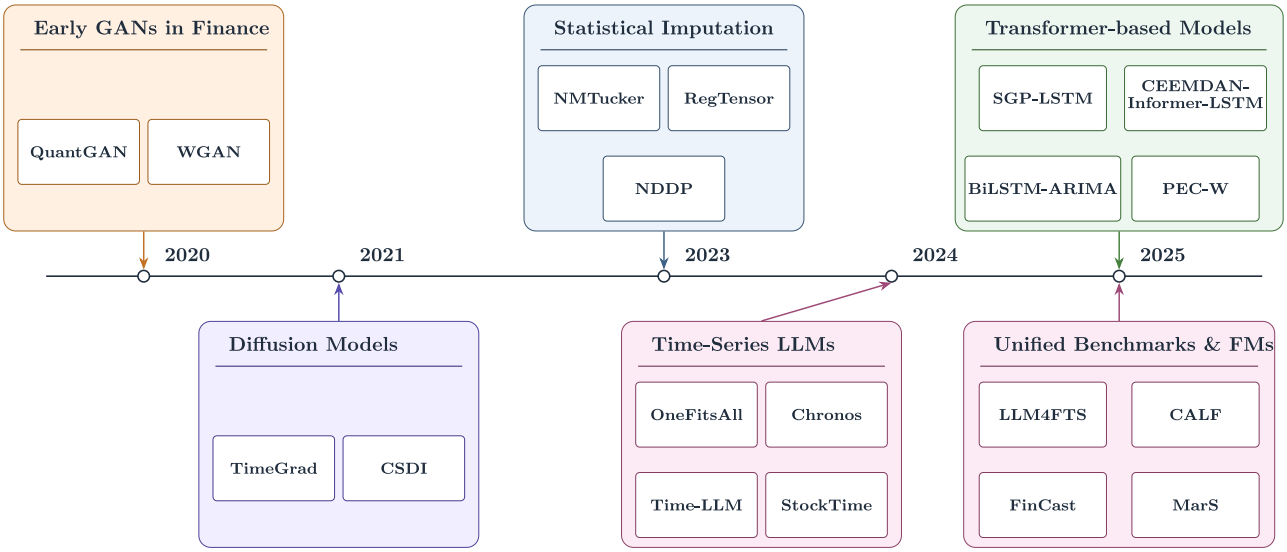


Fig. 1 | Evolution of financial time-series generation techniques over the past decade. Timeline of representative milestones in financial time-series generation (FTSG) from 2020 to early 2026. Colored cards along the central axis denote technical families: coral pink for adversarial generative architectures (GANs), steel blue for diffusion and score-based models, amber orange for time-series foundation models and adapted large language models (LLMs), and sage green for classical statistical, tensor factorization, and deep predictive baselines. Key models include NMTucker, RegTensor, NDDP, PEC-W, SGP-LSTM, TimeGrad, CSDI, Time-LLM, StockTime, CALF, FinCast, and MarS. White circular markers denote calendar years 2020, 2021, 2023, 2024, and 2025; directed arrows align each contribution to its release year.

Table 1 | Comparison of this survey with the six prior surveys cited in the literature comparison

Survey	FTSE	FTSI	FTSS	Eval.	Datasets	# refs	Window
Zhang et al.[21]	✓	×	×	None	Multimodal TS	93	2005–2024
Liang et al.[22]	✓	△	×	None	General TS	123	2017–2024
Kim et al.[23]	✓	×	×	Metrics	General TS	287	2000–2024
Kong et al.[24]	✓	×	×	Metrics	General TS	285	2000–2024
Nie et al.[25]	✓	×	×	Metrics	Financial NLP	318	2001–2024
Arsenault et al.[26]	✓	×	×	Metrics	Financial TS	115	2018–2023
This survey	✓	✓	✓	**Protocol**	**Financial TS**	**131**	**2021–early 2026**

FTSE/FTSI/FTSS columns use ✓/× for whether the sub-task is a first-class topic (△ = mentioned but no dedicated section). Eval. distinguishes a unified *protocol* from per-paper *metrics*. Datasets indicates whether financial benchmarks are inventoried. # refs is the count of references cited in each survey. Window is the publication-year range of each survey's references.

Transformers[17] and large language models (LLMs)[18–20]. These developments, together with their financial adaptations, have produced a fragmented and rapidly expanding body of literature. Zhang et al.[21] survey LLMs for time series, while Liang et al.[22] review time-series foundation models. Kim et al.[23] and Kong et al.[24] review deep time-series forecasting architectures, and Nie et al.[25] and Arsenault et al.[26] review financial-AI applications. These surveys help map parts of the literature, but they leave several task- and evaluation-oriented gaps.

First, the majority of existing surveys remain heavily skewed toward forecasting (extrapolation) and classification, often treating generation merely as an auxiliary data augmentation step rather than a standalone class of problems encompassing synthesis and imputation. Second, the reviews by Zhang et al.[21] and Liang et al.[22] are organized primarily around model families and representation-learning components, whereas the reviews by Kim et al.[23], Kong et al.[24], Nie et al.[25], and Arsenault et al.[26] emphasize architectures or application domains; this organization obscures differences in task formulation, supervision signals, and conditioning contexts. Third, and most critically, existing literature lacks a systematic discussion on evaluation protocols. Assessing a synthetic financial sequence requires moving beyond generic point-wise error metrics (like MSE or MAE) to evaluate cross-sectional dependence, tail-risk preservation, and downstream utility—areas where current surveys offer little standardization or critical synthesis. Finally, there is a scarcity of resource-oriented reviews that comprehensively catalog financial benchmarks, datasets, and the practical biases (e.g., survivorship bias) that plague empirical financial research.

To substantiate these gaps concretely, Table 1 positions our survey against the six prior surveys cited above along five explicit dimensions: task coverage (whether each of the three FTSG sub-tasks—extrapolation (FTSE), imputation (FTSI), and synthesis (FTSS)—is treated as a first-class topic), evaluation treatment (whether a multi-dimensional protocol is provided, beyond per-paper metric tables), dataset cataloging (whether FTSG-relevant financial benchmarks are systematically inventoried), number of references in the bibliography (coded approximately from the available versions), and time window of coverage. Our coding suggests that no prior survey jointly covers all three FTSG sub-tasks with a dedicated evaluation protocol and a curated financial dataset catalog; each gap maps to one of the four motivating critiques above and motivates a corresponding contribution of our survey discussed below.

To address these critical gaps, this paper provides a systematic, evaluation-oriented survey of FTSG, focusing on the intersection of generative AI and financial econometrics over the past five years (2021–early 2026). We elevate the discourse from a mere enumeration of literature to a structured problem formulation and benchmarking perspective. Our contributions are fourfold. First, we propose a task-technique two-level

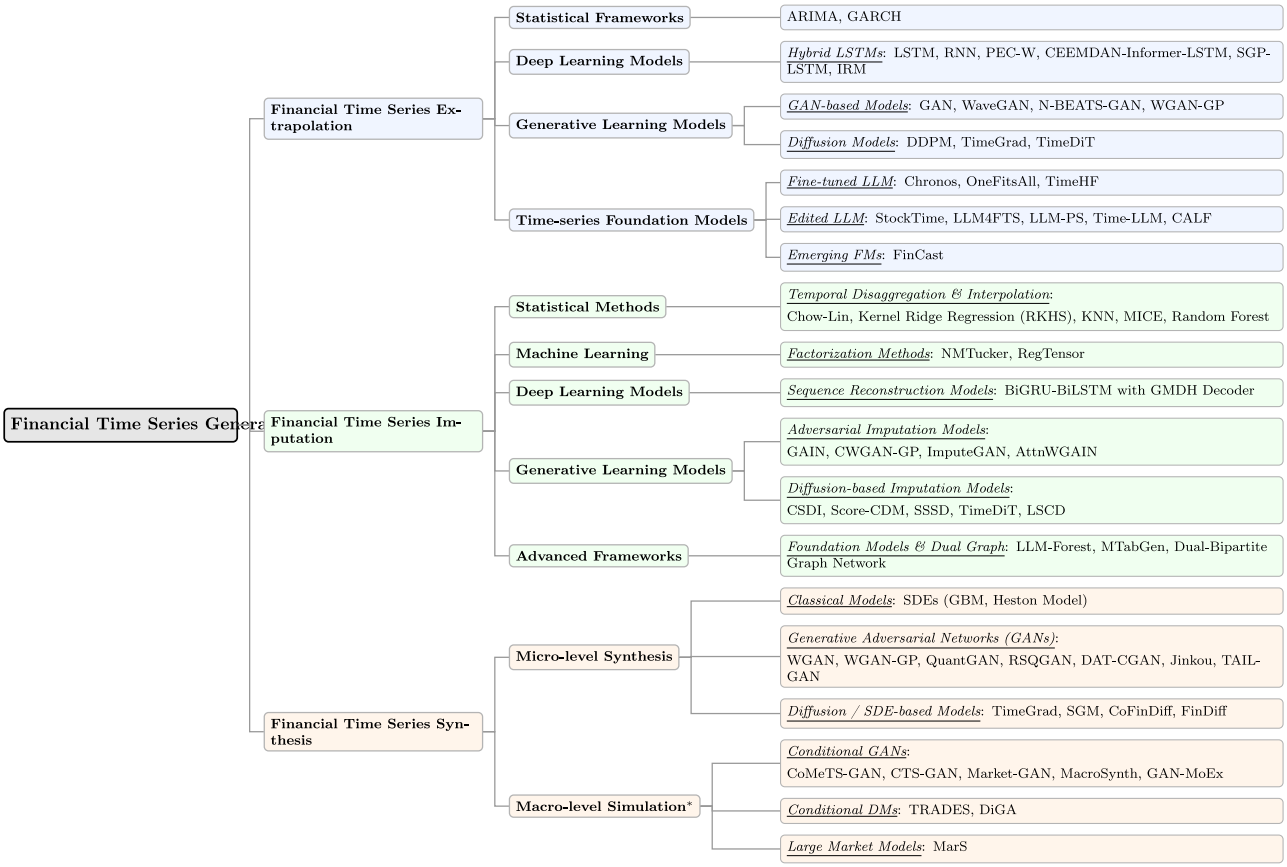


Fig. 2 | Hierarchical two-level taxonomy of financial time-series generation. Systematic classification structure decoupling financial generation tasks from generative modeling paradigms. The primary tree root partitions FTSG into three functional task branches. Light blue shaded blocks designate the forward-looking Financial Time-Series Extrapolation (FTSE) branch, spanning statistical autoregressive models, deep sequential networks, adversarial frameworks, diffusion dynamics, and foundation model adaptations. Light green shaded blocks designate the bidirectional Financial Time-Series Imputation (FTSI) branch, covering temporal disaggregation, tensor factorization baselines, recurrent encoders, generative adversarial imputers, score-based diffusion methods, and graph-augmented transformer architectures. Light orange shaded blocks designate the generative Financial Time-Series Synthesis (FTSS) branch, categorized into micro-level asset return synthesis and macro-level market microstructure simulation across conditional GANs, diffusion models, and large market agent simulators. Grey orthogonal branch lines indicate conceptual inheritance. The asterisk (*) indicates that only purely generative model families are enumerated in the synthesis branch.

taxonomy. By comprehensively reviewing 131 studies published in 2021–early 2026, we develop a hierarchical classification system that decouples the *what* (extrapolation, imputation, and synthesis) from the *how* (statistical, deep discriminative, GAN, diffusion, and foundation-model approaches), enabling a clearer mapping of generative techniques to financial use cases. Second, we establish a systematic evaluation framework for FTSG that moves beyond generic reconstruction metrics to encompass distributional fidelity, stylized-facts preservation, downstream utility, and robustness to regime shifts. Third, we compile and categorize public data resources, APIs, and emerging benchmark suites, including FinTSB[27] and TFB[28], specifically relevant to financial generation, and discuss practical issues in dataset construction. Fourth, we synthesize open challenges in financial validity and identify actionable future directions, including financial-theory-informed generative modeling and causal representation learning.

The remainder of this paper is organized according to the IRDM structure. The Results section presents our core survey findings across six thematic areas: problem formulation and two-level taxonomy, task-specific analyses of extrapolation, imputation, and synthesis, downstream quantitative applications, evaluation protocols for financial realism, and data resources with benchmarks. The Discussion section critically examines pressing open challenges and future research directions, synthesizing the implications for financial generative modeling. Finally, the Methods section details the systematic literature retrieval strategy and PRISMA-compliant screening protocol[29], inclusion criteria, and coding procedures governing this survey.

Results

Problem formulation and setup

To establish a rigorous conceptual foundation, Fig. 2 presents a comprehensive two-level hierarchical taxonomy that structures the entire financial time-series generation landscape across task formulations and algorithmic families. Let $\mathbf{X} = \{x_1, x_2, \ldots, x_T\}$ denote a multivariate financial time series, where each $x_t \in \mathbb{R}^d$ represents a vector of financial variables (e.g., asset prices, returns, trading volumes, or limit order book states) at time step t. Financial time series are inherently stochastic, heavily influenced by non-stationary market regimes, and exhibit complex inter-variable dependencies. The core objective of Financial Time-Series Generation (FTSG) is to learn a probabilistic generative model P_θ capable of producing synthetic sequences $\hat{\mathbf{X}}$ that capture both the marginal distributions and the temporal dynamics of the true data-generating process P_{data}.

Task-level taxonomy

As illustrated in Fig. 3, FTSG comprises three distinct problem formulations based on generation context.

Financial time-series extrapolation is a conditional, forward-looking generation task. Given a historical prefix $\mathbf{x}_{1:t}$, the goal is to generate future trajectories $\mathbf{x}_{t+1:T}$. The predictive distribution is modeled as $P(\mathbf{x}_{t+1:T}|\mathbf{x}_{1:t};\theta) = \prod_{i=t+1}^{T} P(x_i|\mathbf{x}_{1:i-1};\theta)$. This encompasses probabilistic

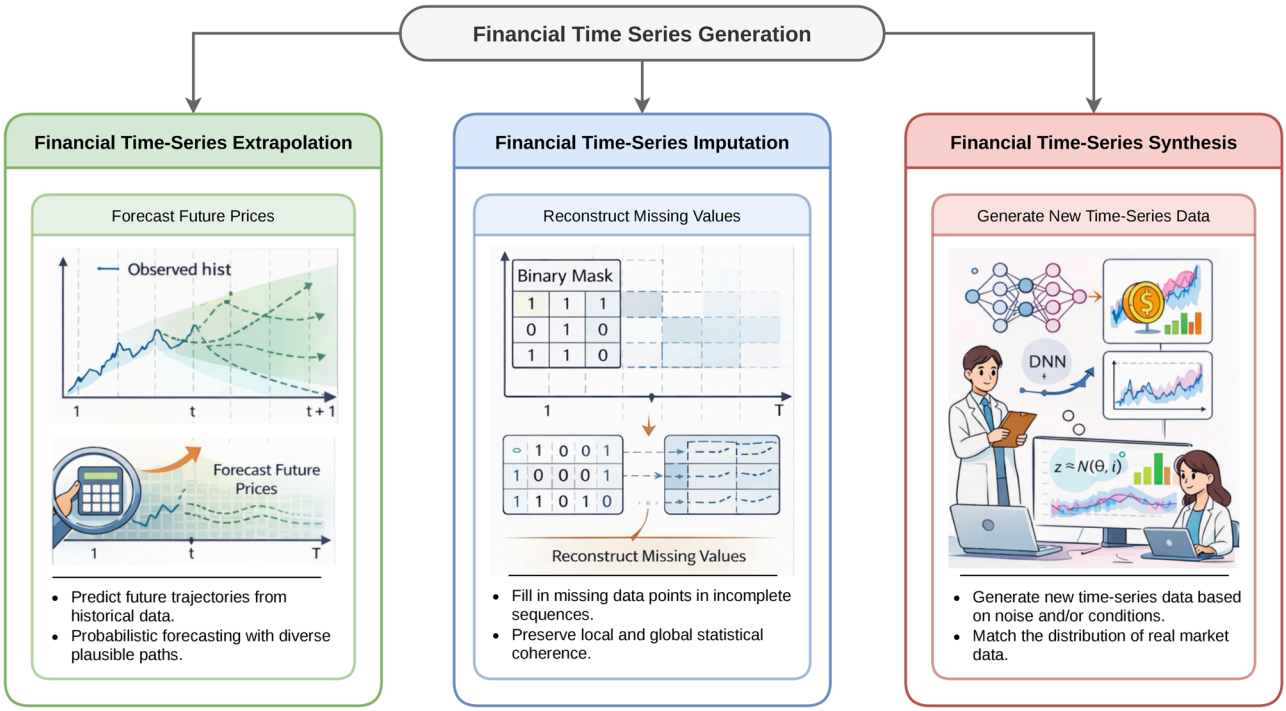


Fig. 3 | Structural comparison of core financial time-series generation tasks. Schematic representation contrasting the conditioning context, temporal boundary conditions, and output targets across three generation formulations. **a** Financial time-series extrapolation (FTSE): conditional forward-looking trajectory continuation, where a past historical prefix (solid line markers) guides probabilistic generation of unobserved future steps (dashed continuation markers). **b** Financial time-series imputation (FTSI): closed-format bidirectional reconstruction, where known observed points bound an interval of missing or corrupted values, reconstructed via temporal and cross-sectional conditioning. **c** Synthetic sequence generation (FTSS): unconditional or attribute-guided manifold creation from latent Gaussian noise and market condition vectors, simulating realistic synthetic price paths and order books without pointwise historical alignment.

forecasting, where the output is a set of plausible future paths rather than a single point estimate.

Financial time-series imputation is a bidirectional, closed-format generation task. Let $\mathbf{M} \in \{0,1\}^T$ be a binary mask indicating missing values. Given an incomplete sequence with missing elements $\mathbf{x}_{miss} = \mathbf{x} \odot \mathbf{M}$ and observed elements $\mathbf{x}_{obs} = \mathbf{x} \odot (1 - \mathbf{M})$, the objective is to reconstruct the missing data while preserving local and global statistical coherence: $P(\mathbf{x}_{miss}|\mathbf{x}_{obs}; \theta)$.

Synthetic sequence generation, or data synthesis, is an unconditional or conditionally guided generation task aimed at creating entirely new data manifolds. The model learns to map a latent noise variable $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$ (and optionally a condition c such as market regime or volatility target) to a realistic sequence: $\hat{\mathbf{X}} = G_\theta(\mathbf{z}, c)$, aiming to minimize the divergence between the generated distribution P_g and the real distribution P_{data}.

Technique-level taxonomy

To map the rapid evolution of algorithmic approaches, the underlying generative techniques can be grouped into six families.

Statistical and probabilistic methods include classical approaches such as ARIMA[30] and GARCH[31], which model conditional mean and variance based on rigid mathematical assumptions.

Discriminative deep sequence models include recurrent neural networks (RNNs)[32] and long short-term memory (LSTM) networks[33] optimized via pointwise loss functions.

Adversarial generative models (GANs)[15] use generator-discriminator dynamics, including the Wasserstein GAN with gradient penalty (WGAN-GP)[34,35], to implicitly learn the data distribution.

Diffusion and score-based models[16,36,37] learn to reverse a gradual noising process, offering stable training and high-fidelity distribution matching.

Foundation models and LLM-based models are recent adaptations of large pre-trained Transformers[17,18] to the time-series domain, represented by models such as Chronos[20], OneFitsAll[38], and Time-LLM[19].

Controllable and multimodal generative models, including TimeWeaver[39], T2S[40], and dual-sourced text–time diffusion models[41], are designed to generate sequences conditioned on textual instructions, macroeconomic indicators, or specific volatility regimes.

The detailed classification of FTSG methods, organized by task and technique, is summarized in Fig. 2 for a clear overview.

Mapping tasks to techniques

Table 2 provides a unified mapping of these tasks to the corresponding techniques, highlighting the evolution from deterministic prediction to flexible, scenario-based generative paradigms. Specifically, for the extrapolation task, the landscape spans from deep sequence architectures such as SGP-LSTM[42] that forecast prices and returns using deterministic metrics like MSE, to advanced diffusion and foundation models such as TimeGrad[13], Chronos[20], and FinCast[43]. These newer paradigms handle multivariate and multi-asset series, shifting the evaluation focus toward proper probabilistic scoring rules such as the continuous ranked probability score (CRPS)[44] and zero-shot capabilities. In the realm of imputation, missing fundamentals and prices are primarily addressed using tensor methods such as RegTensor[45] and diffusion models such as CSDI[11] and TimeDiT[46], where performance is gauged by reconstruction error and temporal coherence. Finally, for data synthesis, generative frameworks such as QuantGAN[8] and CoFinDiff[47] are employed to simulate realistic returns, market scenarios, and complex limit order flows. Consequently, the evaluation criteria for synthesis shift entirely away from point-wise accuracy, prioritizing instead the preservation of stylized facts, distributional fidelity, and the practical downstream utility of the generated data.

Table 2 | Unified task-technique-evaluation mapping for financial time-series generation

Task	Model family	Representative approaches	Data type	Primary evaluation
Extrapolation	Deep Sequence	SGP-LSTM[42]	Prices, Returns	MSE, Directional Accuracy
	Diffusion Models	TimeGrad[13]	Multivariate Series	CRPS, Predictive Likelihood
	Foundation Models	Chronos[20], FinCast[43]	Multi-asset	Zero-shot MSE, MAE
Imputation	Tensor Methods	RegTensor[45]	Fundamentals	Reconstruction Error
	Diffusion Models	CSDI[11], TimeDiT[46]	Missing Prices	Temporal Coherence
Synthesis	GANs	QuantGAN[8]	Returns	Stylized Facts
	Diffusion Models	CoFinDiff[47]	Scenarios	Distributional Fidelity
	Controllable	Market-GAN[142]	Order Flow (LOB)	Downstream Utility

Extrapolation: problem characteristics

Financial Time-Series Extrapolation (FTSE) is formally defined as the task of generating future trajectories $\mathbf{x}_{t+1:T}$ given a historical context $\mathbf{x}_{1:t}$. Unlike traditional point forecasting, modern FTSE is approached as a probabilistic conditional generation problem: learning the conditional distribution $p(\mathbf{x}_{t+1:T}|\mathbf{x}_{1:t})$ to capture the inherent uncertainty and multi-modal nature of financial markets. This shift acknowledges that financial markets are stochastic systems with complex, time-varying dynamics.

Four characteristics distinguish this problem from general sequence extrapolation.

The causality constraint requires strict causality: future values depend solely on past observations without any look-ahead bias. This is analogous to autoregressive language modeling, but with continuous, real-valued tokens.

Non-stationarity means that financial time series exhibit time-varying statistical properties (e.g., changing volatility regimes, structural breaks). The conditional distribution $p(\mathbf{x}_{t+1:T}|\mathbf{x}_{1:t})$ itself evolves over time, requiring models to adapt to distributional shifts.

The low signal-to-noise ratio reflects that market data is dominated by noise, making it challenging to separate genuine predictive signals from random fluctuations. This often leads to overfitting in deterministic models.

High-dimensional dependencies arise in multivariate settings, where assets exhibit complex cross-sectional dependencies (e.g., lead-lag relationships, sector correlations). Modeling these dependencies is crucial for coherent joint generation.

Formally, the objective is to learn a generator G_θ that maps the context $\mathbf{x}_{1:t}$ to a distribution over future sequences, often parameterized as a set of samples $\{\mathbf{x}^{(i)}_{t+1:T}\}_{i=1}^{N}$. The evaluation metrics extend beyond point-wise accuracy (e.g., MSE) to probabilistic scoring rules such as CRPS[44] and the energy score[48], and domain-specific financial utility measures (e.g., portfolio returns, risk metrics).

Extrapolation: classical and hybrid predictive models

Classical statistical frameworks have historically provided the mathematical bedrock for modeling financial time-series dynamics through rigorous probabilistic assumptions. A prominent example is the Auto-Regressive Integrated Moving Average (ARIMA) model[30], which explicitly formulates temporal dependencies by combining autoregressive (AR) processes, differencing (I) to achieve stationarity, and moving average (MA) components. For a given time series y_t, an ARIMA(p, d, q) specification is mathematically expressed as:

$$\nabla^d y_t = c + \sum_{i=1}^{p} \phi_i \nabla^d y_{t-i} + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j} + \varepsilon_t \tag{1}$$

where ∇^d represents the d-th order difference operator, ϕ_i and θ_j are the AR and MA coefficients respectively, and $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ acts as white noise. While ARIMA models are effective at isolating linear relationships, they inherently fail to encompass complex market phenomena such as volatility clustering and heavy-tailed distributions. To capture these time-varying variance dynamics, the generalized autoregressive conditional heteroskedasticity (GARCH) framework[31] was introduced. The conditional variance σ_t^2 in a GARCH(p, q) model is iteratively defined by past squared innovations and past variances:

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^{q} \beta_j \sigma_{t-j}^2 \tag{2}$$

where $\omega > 0$, $\alpha_i \geq 0$, and $\beta_j \geq 0$. Although extensions such as EGARCH[49] further accommodate asymmetric leverage effects resulting from negative shocks, these purely parametric methods struggle to scale to high-dimensional dependencies and intricate non-linear patterns.

To address the constraints of classical statistical models, the field transitioned toward deep learning architectures, particularly recurrent neural networks (RNNs)[32] and long short-term memory (LSTM) networks[33]; recent financial applications include the hybrid models studied by Qin et al.[50], Li et al.[51], Caliskan[52], and Li et al.[42]. LSTMs leverage a sophisticated gating mechanism-comprising forget (f_t), input (i_t), and output (o_t) gates alongside a cell state (C_t) and hidden state (h_t)-to effectively capture long-range temporal dependencies without suffering from severe vanishing gradients:

$$f_t = \sigma(\mathbf{W}_f \cdot [h_{t-1}, x_t] + b_f) \tag{3}$$

$$i_t = \sigma(\mathbf{W}_i \cdot [h_{t-1}, x_t] + b_i) \tag{4}$$

$$\tilde{C}_t = \tanh(\mathbf{W}_C \cdot [h_{t-1}, x_t] + b_C) \tag{5}$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \tag{6}$$

$$o_t = \sigma(\mathbf{W}_o \cdot [h_{t-1}, x_t] + b_o) \tag{7}$$

$$h_t = o_t \odot \tanh(C_t) \tag{8}$$

However, when standard LSTMs are optimized using traditional pointwise loss functions such as the Mean Squared Error (MSE), defined as $\mathcal{L}_{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$, they often overfit the ubiquitous noise present in financial data.

Consequently, recent literature has pivoted towards sophisticated hybrid strategies that couple deep learning backbones with domain-specific signal processing or invariant learning techniques to enhance robustness. Denoising-based hybrids, for example, incorporate explicit signal decomposition to systematically separate underlying predictive trends from stochastic market noise. The PEC-W model[52] leverages the discrete wavelet transform (DWT)[53] to split raw input series into multi-resolution scale and

wavelet representations:

$$x(t) = \sum_k c_{J,k}\phi_{J,k}(t) + \sum_{j=1}^{J}\sum_k d_{j,k}\psi_{j,k}(t) \tag{9}$$

where ϕ and ψ denote the scaling and wavelet functions. These denoised approximation coefficients provide a cleaner input signal for the LSTM, a concept similarly explored by CEEMDAN (Complete Ensemble Empirical Mode Decomposition with Adaptive Noise)[54]-Informer[55]-LSTM model of Li et al.[51], which decomposes the signal into intrinsic mode functions. Beyond denoising, feature engineering hybrids seek to automate the discovery of robust predictive inputs. The SGP-LSTM framework[42] employs symbolic genetic programming[56] to iteratively evolve technical indicators, steering the search via a multi-objective fitness function that balances return, consistency, and predictive correlation (IC):

$$\text{Fitness} = \text{TopR} + \lambda_1 \times \text{Monotonicity} + \lambda_2 \times \text{IC} \tag{10}$$

Finally, invariant learning hybrids target the problem of distributional shifts across varying market regimes. Building on invariant risk minimization (IRM)[57], Cao et al.[58] propose a financial adaptation that enforces the extraction of representations that hold true regardless of whether the market is in a bullish, bearish, or sideways phase, minimizing an objective that penalizes environmental variance:

$$\mathcal{L}_{IRM} = \underbrace{R^e(\mathbf{w}\cdot\Phi)}_{\text{Empirical risk}} + \underbrace{\gamma||\nabla_{\mathbf{w}|_{\mathbf{w}=1.0}} R^e(\mathbf{w}\cdot\Phi)||_2^2}_{\text{Invariant penalty}} \tag{11}$$

where Φ acts as the representation function and $\mathbf{w}$ is a linear classifier. Despite these multifaceted improvements, hybrid predictive architectures ultimately generate deterministic point forecasts, fundamentally limiting their capacity to model the full predictive distribution essential for robust financial scenario generation.

Extrapolation: generative models

Recognizing the intrinsic limitations of deterministic forecasting, the paradigm of generative models shifts the focus toward learning the underlying data distribution $p(\mathbf{x})$ or the conditional probability $p(\mathbf{x}_{t+1:T}|\mathbf{x}_{1:t})$. This transition allows practitioners to sample multiple plausible future trajectories, accommodating the multi-modal uncertainty inherent in financial markets. Early financial studies adapted generative adversarial networks (GANs)[15]; examples include TimeGAN[7] and the conditional GAN proposed by Naritomi and Adachi[9]. These models deploy a generator G and a discriminator D locked in a competitive minimax game. The generator synthesizes sequences from historical context $\mathbf{x}_{1:t}$ and a latent noise vector $\mathbf{z}$, while the discriminator attempts to distinguish real historical samples from the generated distributions:

$$\min_G \max_D \mathbb{E}_{\mathbf{x}\sim p_{data}}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}\sim p_z}[\log(1 - D(G(\mathbf{z})))] \tag{12}$$

While conceptually elegant, GANs can be difficult to train on volatile financial data, frequently suffering from mode collapse and unstable convergence. To mitigate these issues, specialized adaptations have been devised. WaveGAN[59] incorporates a multi-resolution DWT denoising mechanism combined with a Wasserstein GAN equipped with gradient penalty (WGAN-GP)[34, 35]. The WGAN-GP loss enforces a Lipschitz constraint to prevent vanishing gradients:

$$\mathcal{L}_{\text{WGAN-GP}} = \mathbb{E}_{\tilde{\mathbf{x}}\sim p_g}[D(\tilde{\mathbf{x}})] - \mathbb{E}_{\mathbf{x}\sim p_{data}}[D(\mathbf{x})] + \lambda\mathbb{E}_{\hat{\mathbf{x}}\sim p_{\hat{\mathbf{x}}}}[(\|\nabla_{\hat{\mathbf{x}}} D(\hat{\mathbf{x}})\|_2 - 1)^2] \tag{13}$$

where $\hat{\mathbf{x}}$ represents a convex interpolation of real and generated instances. Concurrently, models like N-BEATS-GAN[60] fuse interpretable basis expansions[61] with an adversarial framework, modeling future values as a sum of trend and seasonality bases $\hat{\mathbf{y}} = \sum_{i=1}^{N} \text{Basis}_i(\boldsymbol{\theta}_i)$. While these modifications enhance stability and interpretability, GANs often yield over-smoothed samples that fail to encapsulate the sharp, multi-modal nature of empirical return distributions.

To achieve higher fidelity in distribution matching, Denoising Diffusion Probabilistic Models (DDPMs)[16] have emerged as a more stable and expressive alternative. Diffusion frameworks systematically destroy structured data by injecting Gaussian noise over T forward steps-$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$-and then learn a parameterized reverse process $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \sigma_t^2\mathbf{I})$ to reconstruct the signal. The optimization simplifies to predicting the incrementally added noise $\boldsymbol{\epsilon}$:

$$\mathcal{L}_{simple} = \mathbb{E}_{t,\mathbf{x}_0,\boldsymbol{\epsilon}}[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\|^2] \tag{14}$$

For context-conditioned financial generation, TimeGrad[13] integrates an RNN-based encoder with a score-based diffusion decoder, generating future states via Langevin dynamics where the context encoding $\mathbf{c}$ steers the reverse sampling:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{c})\right) + \sigma_t\mathbf{z} \tag{15}$$

Addressing the limitations of recurrent architectures in capturing extended dependencies, TimeDiT[46] replaces the RNN with a diffusion Transformer of the type introduced by Peebles and Xie[62]. Utilizing patch embeddings and multi-head attention, TimeDiT operates on a μ-parameterized loss function, explicitly aligning the predicted mean of the reverse process with its theoretical target:

$$\mathcal{L} = \sum_{t=1}^{T} \mathbb{E}_{q(\mathbf{x}_t^{tar}|\mathbf{x}_0^{con})}\left[\|\boldsymbol{\mu}(\mathbf{x}_t^{tar}, \mathbf{x}_0^{con}) - \boldsymbol{\mu}_\theta(\mathbf{x}_t^{tar}, t|\mathbf{x}_0^{con})\|^2\right] \tag{16}$$

An alternative diffusion formulation leverages continuous-time score matching[36,37], optimizing a score network to approximate the gradient of the log-density:

$$\mathcal{L}_{score} = \mathbb{E}_{t,\mathbf{x}_0,\boldsymbol{\epsilon}}\left[\left\|\boldsymbol{\epsilon} - \sigma_t\nabla_{\mathbf{x}_t}\log p_\theta(\mathbf{x}_t)\right\|^2\right] \tag{17}$$

While diffusion and score-based models generate high-fidelity distributions with sharp multi-modal properties, their iterative sampling procedures introduce significant inference latency, presenting an ongoing challenge for high-frequency financial applications.

Extrapolation: time-series foundation models

Inspired by the success of Transformers[17] and large language models[18] in natural language processing, current research is heavily invested in adapting the "pre-train, then fine-tune" paradigm to financial time-series extrapolation. These foundation models are trained on massive, heterogeneous datasets to derive generalized temporal representations, effectively enabling zero-shot inference and cross-domain adaptation. The primary challenge in this domain is bridging the continuous numerical space of financial data with the discrete, tokenized vocabulary natively processed by Transformer-based language models. Tokenization-based fine-tuning strategies directly address this by casting forecasting as a next-token prediction task. For instance, Chronos[20] scales time-series observations and maps them into a fixed dictionary of tokens using quantile binning:

$$\text{token}(\mathbf{x}) = \arg\min_i |\text{x} - \text{q}_\text{i}| \tag{18}$$

where q_i represent pre-computed quantiles. While this approach benefits from robust zero-shot generalization, the inherent discretization process sacrifices fine-grained numerical precision. Conversely, patch-based fine-tuning techniques, as exemplified by OneFitsAll[38], bypass discretization by

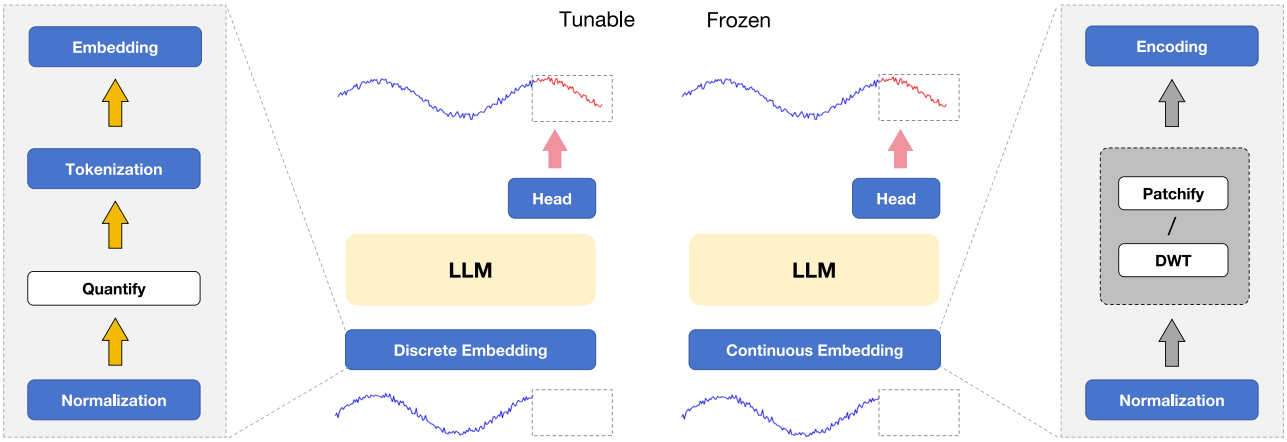


Fig. 4 | Architectural paradigms for adapting foundation models to financial time series. Schematic diagram contrasting tokenization- and patch-based fine-tuning against model editing and prompt reprogramming. **a** Fine-tuning paradigm (left): numerical continuous time series are transformed into discrete token vocabularies via quantile binning or linear patch projection layers, followed by back-propagation through transformer backbones. **b** Prompt reprogramming and cross-modal model editing paradigm (right): a frozen foundation model backbone receives auxiliary steering signals from textual market prompts, cross-attention alignment bridges, and lightweight adapter modules, avoiding full-parameter gradient updates while preserving numerical precision. Blue rectangular boxes represent time-series input vectors, purple layered blocks denote multi-head self-attention transformer layers, and green highlighted modules represent learnable cross-modal projection heads.

segmenting the time-series into sequences of sub-series patches. Each patch is projected into the high-dimensional embedding space of the LLM via a linear transformation:

$$\mathbf{E}_{patch} = \mathbf{W}_p \mathbf{x}_{patch} + \mathbf{b}_p \tag{19}$$

These continuous patch embeddings are then prepended as visual-like tokens, preserving numerical fidelity while leveraging the LLM's cross-patch attention mechanisms.

As illustrated in Fig. 4, beyond conventional fine-tuning, sophisticated model editing and reprogramming approaches have emerged to align disparate modalities using lightweight adapters, averting the computational burden of full-parameter updates. StockTime[63] enriches time-series representations by encoding high-frequency tick data alongside handcrafted textual summaries (e.g., "bullish engulfing pattern") via a dual-encoder. Similarly, frameworks like LLM4FTS[64] and LLM-PS[65] use dynamic time warping (DTW)[66] and text-translation modules to bridge the semantic divide between raw price trajectories and natural language descriptions. A more fundamental alignment is achieved through Prompt-as-Prefix reprogramming, popularized by Time-LLM[19]. This technique steers frozen LLMs by projecting time-series patches onto a set of learned "text prototypes" $\mathbf{P} \in \mathbb{R}^{M \times d}$ using cross-attention dynamics:

$$\alpha_{ij} = \text{softmax}\left(\frac{(\mathbf{W}_q \mathbf{x}_i) \cdot (\mathbf{W}_k \mathbf{p}_j)^\top}{\sqrt{d}}\right) \tag{20}$$

$$\tilde{\mathbf{x}}_i = \sum_j \alpha_{ij} \mathbf{p}_j \tag{21}$$

The resulting reprogrammed patches act as informative prefixes for the frozen language model. Expanding upon this, the CALF architecture[67] employs a dual-branch structure with consistency losses to penalize the discrepancy between textual and temporal feature distributions.

In parallel, researchers are developing purpose-built forecasting architectures that integrate domain-specific inductive biases directly into the foundation model backbone. TimeHF[68] introduces a time-series policy optimization (TSPO) framework that aligns generative outputs with specific financial human preferences, such as maximizing risk-adjusted returns, using reinforcement learning from human feedback (RLHF)[69,70]:

$$\mathcal{L}_{TSPO} = -\mathbb{E}_{\tau \sim \pi_\theta}[R(\tau)] + \beta\, \mathrm{KL}\left[\pi_\theta \parallel \pi_{\text{ref}}\right] \tag{22}$$

where π_θ denotes the policy (generative model) parameterized by θ, $\tau \sim \pi_\theta$ is a sampled generation trajectory, $R(\tau)$ is the reward function that scores trajectory quality according to human financial preferences, β controls the strength of the KL-divergence regularization, and π_{ref} is the frozen reference policy that prevents catastrophic drift from pre-trained behavior. FinCast[43] addresses the multi-scale nature of market data using a sparse mixture-of-experts (MoE) backbone[71,72] enhanced with learnable frequency embeddings. To ensure the model delivers both accurate point estimates and reliable probabilistic bounds, it optimizes a joint Point-Quantile loss:

$$\mathcal{L}_{PQ} = \lambda_1 \mathcal{L}_{point} + \lambda_2 \mathcal{L}_{quantile} \tag{23}$$

$$\mathcal{L}_{quantile} = \sum_q \rho_q(y - \hat{y}_q), \quad \rho_q(u) = u(q - \mathbb{I}(u<0)) \tag{24}$$

where $\mathcal{L}_{point}$ handles deterministic accuracy and the pinball loss derived from quantile regression[73] governs risk coverage, with the reported aim of improving robust zero-shot performance across asset classes.

Extrapolation: emerging trends

Several bottlenecks remain before generative models can be operationalized in live financial systems. A primary objective is the development of unified hybrid architectures that can combine the low-latency inference speeds of discriminative models with the distributional robustness of diffusion frameworks, potentially leveraging knowledge distillation[74]. Evaluation protocols also need to be expanded; existing metrics such as the continuous ranked probability score (CRPS) should be augmented with practical financial utility measures, including risk-adjusted returns based on measures such as the Sharpe ratio[75] and maximum drawdowns, so that models better reflect practical trading objectives. Furthermore, future iterations of generative models should incorporate causal constraints that respect market microstructures[76,77], reducing the look-ahead biases that can affect synthetic market simulators. Finally, advancing computationally efficient fine-tuning protocols is important to help ensure that massive foundation models can rapidly adapt to sudden structural regime shifts without succumbing to catastrophic forgetting[78,79].

Extrapolation: research synthesis

The progressive evolution of financial time-series extrapolation models demonstrates a clear overarching trajectory in machine learning: transitioning from deterministic, discriminative point predictors to expressive generative distribution learners, and ultimately culminating in large-scale foundation models. As synthesized in Table 3, each methodological paradigm presents distinct trade-offs between inferential speed, distributional fidelity, and generalization capability. Classical and hybrid predictive models, such as ARIMA and LSTM-based architectures, remain attractive in latency-sensitive environments like high-frequency execution algorithms, where rapid point forecasting is important. However, their structural inability to quantify tail-risk and multimodal uncertainty limits their utility in comprehensive risk management. To resolve this, generative architectures-notably GANs and diffusion models-are often used in scenario generation and stress testing by synthesizing plausible multi-modal market trajectories. Studies of TimeGrad, TimeDiT, and related financial diffusion models often report better calibrated distributions than selected GAN baselines[13, 46, 47], although this gain typically comes at the cost of substantially higher computational burden during iterative reverse-sampling. Time-series foundation models have recently become an important direction by exhibiting strong zero-shot forecasting capabilities in benchmark experiments, enabling the deployment of predictive systems across previously unseen assets or shifting market regimes. Despite this progress, foundation models still face with persistent challenges regarding temporal alignment, cross-modal semantic gaps, and the propensity to hallucinate implausible financial patterns over extended horizons.

Imputation: problem formulation and characteristics

Financial time-series imputation is fundamentally formulated as a bidirectional, closed-format generation task. Given a multivariate sequence $\mathbf{X} \in \mathbb{R}^{T\times d}$ and a corresponding binary indicator mask $\mathbf{M} \in \{0, 1\}^{T\times d}$ where $m_{t,i} = 1$ denotes a missing value, the objective is to learn a conditional distribution $p(\mathbf{X}_{miss}|\mathbf{X}_{obs}; \theta)$ to accurately reconstruct the missing elements $\mathbf{X}_{miss} = \mathbf{X} \odot \mathbf{M}$ conditioned strictly on the observed elements $\mathbf{X}_{obs} = \mathbf{X} \odot (\mathbf{1} - \mathbf{M})$.

In financial contexts, the missingness mechanisms-missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR), formalized by Rubin[80]-present unique challenges. MNAR is particularly prevalent due to structural market events such as sudden trading halts during market crashes or asynchronous trading hours across global exchanges, a setting related to non-synchronous observation of multivariate financial processes[81]. Consequently, imputation models must conquer two primary domain-specific hurdles. First, models must manage *contiguous missingness*, where extended data outages corrupt long temporal windows. Second, models must maintain strict *cross-variable consistency* without introducing look-ahead bias, which is critical to ensure that historical backtests of trading strategies remain structurally valid.

Imputation: statistical, interpolation, and machine-learning baselines

Early methodologies treated financial imputation primarily as a matrix-completion[82] or interpolation problem rooted in explicit statistical assumptions. The classic Chow-Lin method[83] remains a foundational statistical approach for temporal disaggregation, frequently integrated with LASSO regression[84] to estimate high-frequency (e.g., monthly) values from low-frequency (e.g., quarterly) observations using auxiliary market indicators. Later, machine-learning techniques such as *k*-nearest-neighbour imputation[85], multiple imputation by chained equations (MICE)[86], and Random Forests[87], including missForest[88], offered flexible, non-parametric alternatives for tabular data imputation. The heterogeneous-data adversarial learning network proposed for FinTech applications by Khuwaja et al.[89] is included here as a related data-driven baseline. Operating in a continuous functional space, Kernel ridge regression[90, 91], in a fixed-income application[92], estimates missing discount curves by projecting noisy bond prices into a reproducing kernel Hilbert space (RKHS). This framework

Table 3 | Comprehensive comparison of financial time-series extrapolation paradigms

Paradigm	Output format	Key strengths	Primary limitations	Typical financial use cases
Predictive Models	Deterministic point forecasts	Extremely fast inference; highly interpretable methodologies.	Lacks robust uncertainty quantification; fails to model tail risks.	High-frequency trading; algorithmic order execution.
GANs	Sampled stochastic trajectories	Produces sharp, multi-modal sequences without iterative sampling.	Prone to severe training instability and mode collapse.	Scenario generation; counterfactual stress testing.
Diffusion Models	Sampled stochastic trajectories	Achieves strong distributional fidelity and stable convergence.	Iterative sampling process induces prohibitive computational latency.	Risk management; complex portfolio optimization.
Foundation Models	Zero-shot probabilistic forecasts	Exhibits promising cross-domain generalization and rapid adaptability.	Suffers from potential hallucination and difficult text-series data alignment.	Dynamic asset allocation; broad market backtesting.

minimizes a weighted least-squares objective regularized with exponential weighting to deliberately suppress wild extrapolations during periods of high volatility.

As datasets expanded, traditional machine learning models such as *k*-nearest neighbours[85], MICE[86], and Random Forest/missForest imputation[87,88] were widely adopted. Random Forest imputation, in particular, leverages ensemble decision trees to dynamically predict missing values by capturing complex, non-linear feature interactions. Evidence from corporate-distress prediction indicates that tree-based ensembles can remain competitive with, and in some sparse-data settings outperform, deep learners[93]; the precise magnitude of the advantage is dataset-dependent.

Imputation: tensor factorization and deep sequential approaches

To accommodate higher-dimensional non-linearities and multivariate temporal dependencies, research progressed toward deep interpolation and tensor decomposition networks. Deep sequential fusion architectures, such as the BiGRU-BiLSTM framework proposed by Wang et al.[94], capture bidirectional temporal flows using gated recurrent units[95] within a bidirectional recurrent architecture[96], and employ a group method of data handling (GMDH) polynomial decoder[97] to reconstruct complex missing patterns in multivariate series.

Concurrently, tensor factorization models emerged as highly efficient solutions for structural missingness. NMTucker[98], building on Tucker decomposition[99], introduced a non-linear Matryoshka Tucker decomposition, systematically replacing linear core operations with neural network-based reconstructions and enforcing recursive core constraints to mitigate overfitting on sparse financial tensors. Expanding upon this, RegTensor[45] facilitates fast, non-linear coupled tensor completion across multiple sparse datasets. By sharing factor matrices across tensors, it captures latent cross-market dynamics while strictly enforcing uncorrelated temporal embeddings via an orthogonal regularization penalty:

$$\mathcal{L}_{RegTensor} = \mathcal{L}_{recon} + \lambda \parallel \mathbf{U}^{\top}\mathbf{U} - \mathbf{I}\|_F^2 \tag{25}$$

where $\mathbf{U}$ represents the temporal factor matrix and λ controls the orthogonal constraint. While computationally efficient, these models occasionally fail to capture the high-frequency volatility jumps inherent to tick-level financial data.

Imputation: adversarial and generative imputation dynamics

To overcome the smoothing effects typical of standard interpolators, Generative adversarial networks (GANs)[10,15] formulate missing data recovery as a competitive game. However, standard GANs frequently suffer from severe distributional incoherence-commonly referred to as "drift"-during contiguous data outages. To directly address this, AttnWGAIN[100] substitutes traditional convolutional layers with diagonal masked attention blocks, explicitly forcing the generator to jointly learn temporal and feature-wise correlations without accessing masked future tokens. Similarly, ImputeGAN[101] leverages an Informer-based generator[55] utilizing probabilistic sparse attention, significantly reducing the $O(L^2)$ computational complexity of long-sequence dependencies while maintaining fidelity.

To rigorously enforce distributional stability across extensive missing windows, the CWGAIN-GP framework[12] employs a conditional Wasserstein GAN fortified with a gradient penalty[34,35]. By conditioning the adversarial game on the observation mask $\mathbf{M}$ and enforcing Lipschitz continuity, it successfully bounds the generative drift, ensuring that the synthetic interpolations mathematically align with the heavy-tailed distributions of surrounding observable financial regimes.

Imputation: diffusion and foundation models for missing data

Score-based diffusion models[16,36,37] have become a leading approach for probabilistic imputation by treating missing value reconstruction as a conditional generative process. These architectures define a forward process that adds Gaussian noise strictly to the missing regions (or the entire sequence, masked subsequently), and optimize a reverse denoising step conditioned on $\mathbf{X}_{obs}$:

$$\mathcal{L}_{diff} = \mathbb{E}_{\mathbf{x}_0,\boldsymbol{\epsilon},t}[\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{X}_{obs})\|^2] \tag{26}$$

A central architecture in this domain is CSDI[11], which utilizes a dual 2D-attention mechanism to seamlessly integrate temporal dynamics and cross-feature correlations. Subsequent works such as LSSDM, Score-CDM, and SaSDim[102–104] further advanced this paradigm. To eliminate the "boundary disharmony" problem-unrealistic volatility jumps at the borders of observed and imputed regions-advanced diffusion models incorporate explicit consistency constraints across temporal and spatial horizons[105], as well as cross-variable boundaries[106].

Furthermore, models like LSCD[107] and SPDM[108] employ frequency-domain transformations to map 1D time series into 2D periodic representations. This isolates stable macroeconomic cyclical components from high-frequency market microstructure noise during the denoising phase. Expanding this to transformer scales, TimeDiT[46] introduces a diffusion Transformer (DiT)[62] utilizing a unified masking strategy, uniquely enabling fine-tuning-free model editing. This allows quantitative analysts to directly inject external physical constraints or market heuristics into the sampling trajectory.

Simultaneously, large language models (LLMs), building on few-shot language-model capabilities[18], have been adapted for heterogeneous tabular and sequential data imputation. LLM-Forest[109] pioneers an ensemble approach utilizing graph-augmented prompts. It constructs bipartite graphs to map data entries to feature values, utilizing hierarchical merging to retrieve nearest neighbors. These neighbors serve as few-shot prompts for a forest of LLMs, resolving the final imputed value via confidence-weighted voting. Additionally, frameworks like MTabGen[110] bridge diffusion dynamics and mixed-type tabular formats by running parallel forward processes for categorical and continuous numerical features, unifying them via columnar embeddings.

Imputation: emerging trends

Recent literature indicates a substantial shift toward handling complex, non-random missing patterns through specialized extensions of these generative frameworks. As documented in recent methodological surveys by Shan et al.[111] and Du et al.[112], the field is rapidly moving away from isolated point-interpolators toward unified, context-aware sequence modelers. We highlight three critical emerging trends driving this evolution.

The first trend concerns unified multi-task architectures: instead of treating imputation as an isolated preprocessing step, modern models increasingly solve related financial tasks simultaneously to improve representational learning. For example, Miao et al.[113] propose coupled diffusion models that jointly perform forecasting and imputation, leveraging shared latent representations to enforce temporal consistency. To enhance structural robustness, Zhang et al.[114] use dual-attention networks to simultaneously capture local and global dependencies, while Cai et al.[115] employ temporal graph networks (TGNs)[116] to map dynamic cross-asset spillovers during fragmented market outages[117]. Furthermore, Zhang et al.[118] extend this paradigm beyond numerical data, advancing multi-modal frameworks that integrate textual sentiment to inform the imputation of missing price data.

The second trend focuses on high-frequency and microstructure imputation: it centers on the granular complexities of high-frequency trading (HFT) and Limit Order Book (LOB) data. Standard daily-level imputers often fail at the microstructural level; thus, Bamford et al.[119] and Shankar et al.[120] have proposed specialized deep learning architectures and benchmark suites that strictly respect LOB queue dynamics and volume imbalances. To bridge the gap between generative fidelity and the stringent latency requirements of HFT systems, Nater and Khayati[121] introduce scalable probabilistic imputation techniques that bypass the computational overhead of iterative diffusion without sacrificing distributional shape.

The third trend concerns regime-awareness and stress-testing protocols: the evaluation of imputation models has matured to account for market

Table 4 | Comparative analysis of financial time-series imputation frameworks

Methodology	Core mechanism	Key advantages	Primary limitations
Matrix/Tensor Completion	Low-rank factorization	Highly computationally efficient; effective on sparse data.	Fails to capture non-linear volatility clustering.
Tree-Based (Random Forest)	Non-linear ensemble regression	Extremely robust to high missing rates (e.g., 50%+).	Struggles with high-frequency temporal dependencies.
GAN-Based Models	Adversarial distribution matching	Sharp interpolations; mitigates drift via gradient penalties.	Mode collapse; unstable training on extreme outages.
Diffusion Models	Conditional reverse denoising	State-of-the-art fidelity; preserves cross-asset correlations.	High inference latency prohibitive for real-time systems.
LLM/Graph Frameworks	Bipartite graphs & textual prompting	Excels on mixed-type (categorical/numerical) tabular data.	Heavy computational footprint; context-window limits.

non-stationarity. Recognizing that structural breaks often coincide with missing data (e.g., trading halts), Qian et al.[122] and Gao et al.[123] introduce dynamic, regime-aware score-based models specifically designed to adapt to shifting market volatility. Crucially, a growing body of work advocates for *stratified stress-testing evaluation protocols*. Wi et al.[124] emphasize that imputation fidelity must be validated during simulated market crashes rather than strictly during stationary periods, ensuring that imputed data maintains structural integrity when risk managers need it most. Complementing this evaluation, Wang et al.[125] investigate the pervasive issue of look-ahead bias in deep learning imputers, establishing strict cross-validation guidelines to ensure that reconstructed historical sequences do not inadvertently leak future market information into backtesting pipelines.

Imputation: research synthesis

The landscape of financial missing data recovery is defined by a rigorous trade-off between computational efficiency and generative fidelity, as synthesized in Table 4. The empirical case for diffusion-based imputation is, however, less settled than the tabular taxonomy above might suggest. The most-cited diffusion imputer, CSDI[11], reports a 40–65% improvement over probabilistic baselines and a 5–20% improvement over deterministic baselines—but on *healthcare and environmental* datasets, not financial. Direct head-to-head benchmarks that put diffusion imputers (CSDI, TimeDiT, LSCD) on the same financial dataset as GAN imputers (GAIN, AttnWGAIN, ImputeGAN, CWGAIN-GP) are not standard in the literature, and the few within-paper comparisons that exist (e.g. Score-CDM vs GAIN on air-quality or electricity data) do not transfer cleanly to the financial regime of contiguous missingness and regime-switching volatility. The honest reading is therefore: diffusion models are the most promising current direction for high-fidelity imputation under contiguous missingness, but the claim that they "stand unrivaled" for financial imputation specifically rests on architectural plausibility and on transferred evidence rather than on a published financial benchmark. Tensor factorization and tree-based methods remain preferable when the constraint is large-scale historical-database throughput rather than peak reconstruction quality.

Synthesis: problem formulation and characteristics

Financial Time-Series Synthesis (FTSS) refers to the systematic, algorithmic generation of statistically credible and temporally coherent synthetic sequences. The overarching objective is to enlarge or enrich the empirical sample space upon which downstream quantitative models and trading agents are trained. Unlike extrapolation (which predicts the continuation of a partial series) or imputation (which mathematically restores broken or missing intervals), FTSS is designed to create entirely new, unobserved data manifolds from scratch. Formally, the goal is to learn a generative model $p_\theta(\mathbf{X})$ that closely approximates the true multidimensional data-generating distribution $p_{data}(\mathbf{X})$, enabling unconditional or condition-guided sampling of synthetic market trajectories $\hat{\mathbf{X}}$.

The primary functions of FTSS include restoring population heterogeneity, rebalancing extremely heavy-tailed historical episodes to correct dataset skewness, and providing controlled, counterfactual market regimes for rigorous stress testing. As shown in Fig. 5, methodologically, the literature can be categorized across two distinct scales: *micro-level synthesis*, which focuses on generating univariate or low-dimensional asset prices and return sequences, and *macro-level market simulation*, which attempts to reproduce complex, asynchronous market microstructures[76,77] such as order flows and full limit order books (LOB).

Synthesis: micro-level asset returns and pricing dynamics

Micro-level synthesis targets the generation of asset price trajectories and return sequences, serving primarily as a sophisticated data augmentation mechanism. Early applications of generative adversarial networks (GANs)[15] in this domain frequently suffered from severe training instability and mode collapse, largely due to the financial stylized facts[14,126], such as pronounced volatility clustering and non-Gaussian heavy tails.

To overcome these barriers, the field rapidly transitioned to Wasserstein GANs (WGANs)[9,34], which replace the structurally unstable Jensen-Shannon divergence with the Wasserstein-1 (Earth Mover's) distance. The optimization objective focuses on minimizing:

$$W_1(\mathbb{P}_r, \mathbb{P}_g) = \inf_{\gamma \in (\mathbb{P}_r, \mathbb{P}_g)} \mathbb{E}_{(\mathbf{x},\tilde{\mathbf{x}})\sim\gamma}[\| \mathbf{x} - \tilde{\mathbf{x}} \|] \tag{27}$$

where $(\mathbb{P}_r, \mathbb{P}_g)$ denotes the set of all joint distributions mapping the real distribution $\mathbb{P}_r$ to the generated distribution $\mathbb{P}_g$. The WGAN family was extended for finance in several methodologically distinct directions, which we now unpack critically.

QuantGAN[8] established the WGAN+temporal convolutional network (TCN)[127] template for multi-step unconditional price-path generation. Its verification protocol relies on reproducing eight classical stylized facts (leverage, volatility clustering, heavy tails, absence of autocorrelation in returns, long-memory in absolute returns, etc.) on synthetic S&P 500 paths; the same study reports that QuantGAN-generated trajectories pass all eight checks while the baseline convolutional GAN fails on at least three. QuantGAN is, however, restricted to *univariate* return series—a limitation that every constraint-aware variant below explicitly tries to relax.

RSQGAN[128] injects *regime awareness* into the generator by conditioning on a latent regime variable (typically inferred via a hidden Markov model[129,130] over historical returns). The reported value proposition is distributional alignment *across* regimes rather than within a single regime: paths sampled under the bull regime should resemble bull-period historical paths, and likewise for bear and transition periods. The conditioning mechanism is comparatively cheap (a single discrete regime index appended to the noise input) and is the only one of the four that does not require a custom loss function.

DAT-CGAN[131] reformulates the GAN objective around *decision-related* quantities rather than marginal distributions. Concretely, the discriminator is trained against a multi-Wasserstein loss evaluated on portfolio-level statistics (e.g. cumulative PnL, Sharpe ratio[75], drawdown) computed on rolling windows of the generated path. The authors demonstrate on a multi-time-step portfolio-choice task that DAT-CGAN

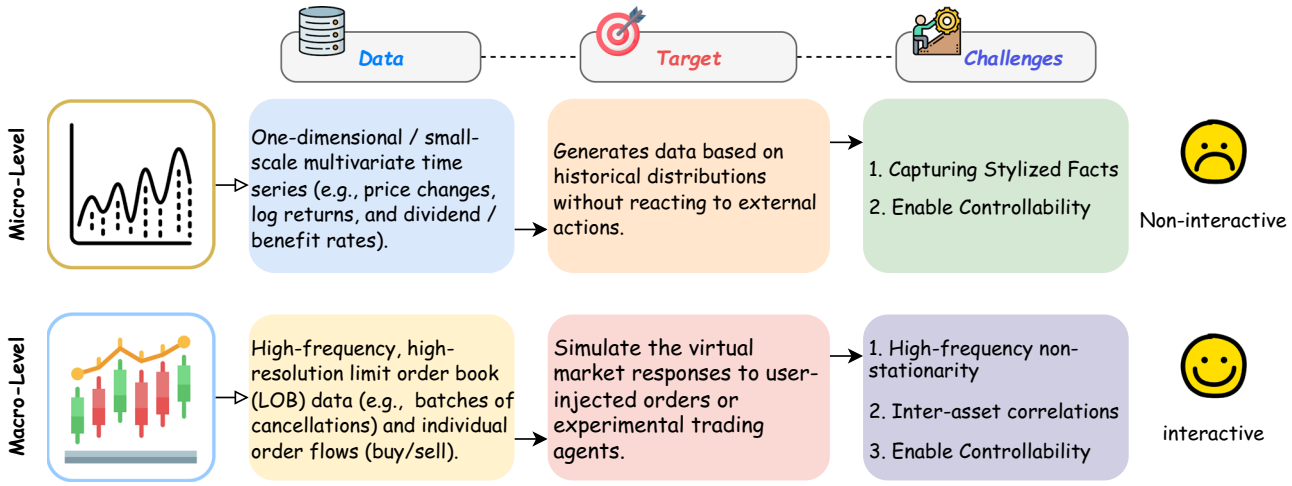


Fig. 5 | Multi-scale architectural paradigms for financial time-series synthesis. Conceptual comparison of synthetic sequence generation partitioned into micro-level return modeling and macro-level market simulation. **a** Micro-level synthesis (left): univariate or low-dimensional asset return path generation via Wasserstein GANs with temporal convolutional backbones and score-matching diffusion models, engineered to preserve stylized econometric facts such as heavy tails, absence of linear autocorrelation, and volatility clustering. **b** Macro-level market simulation (right): multi-agent and limit order book (LOB) queue modeling where generative architectures simulate asynchronous order message flows, bid-ask depth ladders, cross-asset spillover correlations, and macroeconomic counterfactual shock scenarios under regulatory and privacy constraints. Color shading highlights the transition from low-dimensional statistical moments to high-dimensional interactive market topologies.

improves generative quality on both the underlying data and on decision-related quantities relative to strong GAN baselines. The cost is a loss function that is harder to optimize: training stability degrades when the decision-statistic set grows.

Jinkou[132] imposes *state-variable constraints* that keep generated trajectories consistent with economically meaningful drivers (e.g. macro factors, implied volatility surfaces). Unlike the other three, Jinkou is positioned for scenario generation rather than return-path generation, which makes it a natural bridge from micro-level synthesis to macro-level market simulation.

TAIL-GAN[133] stands apart from the other three because its training criterion is not a divergence over the data distribution at all but a *risk-functional score* based on the joint elicitability of Value-at-Risk (VaR) and Expected Shortfall (ES)[134, 135]. The discriminator is optimized against the Acerbi–Szekely scoring function[135], which is the standard backtesting score for VaR/ES; the generator therefore receives gradient signal dominated by tail outcomes rather than by typical samples. This matters because, as Cont et al.[133] argue explicitly, Wasserstein- or cross-entropy-based losses are sample-dominated and structurally fail to capture tail events—which is exactly the regime regulators care about. Their empirical evaluation compares Tail-GAN against a Historical Simulation baseline and a plain WGAN on a five-name universe (AAPL, AMZN, GOOG, JPM, QQQ) for the period 2019–11 to 2019–12. On out-of-sample VaR/ES estimation for static buy-and-hold, mean-reversion, and trend-following strategies, the reported OOS relative errors are: Historical Simulation ~ 10.4%, WGAN ~ 26.9%, Tail-GAN ~ 10.1%—i.e. Tail-GAN closes most of the gap to the oracle, while a vanilla WGAN does roughly 2.5 × worse. On the more demanding score-based test of distributional fidelity under dynamic strategies, Tail-GAN achieves a 21.3% rejection rate versus WGAN's 11.4% and historical simulation's 0%—the higher rejection rate being preferable, since it indicates the synthetic distribution is distinguishable from the test set in the way real markets are. Two limitations temper the result: (i) the headline numbers are on a one-month training window with five names, so scalability to a full S&P 500 universe required eigenportfolio-based dimensionality reduction, with Tail-GAN(Eig) at 25.6% OOS error versus Historical Simulation at 25.9% (a much smaller gap); (ii) Tail-GAN does *not* target stylized-fact reproduction—its evaluation is risk-functional only, so it is complementary to rather than substitutable for QuantGAN/RSQGAN.

Despite the improvements brought by WGANs, score-based diffusion models[36, 37] have been increasingly applied to multi-asset return synthesis with the explicit goal of preserving cross-sectional correlation matrices. The empirical case for this claim is, however, mostly *within-paper* rather than cross-paper: CoFinDiff[47], Fin-DDPM[136], and GBM-Diff[137] each report better cross-correlation preservation against their own GAN baselines on their own chosen datasets, but a shared benchmark that pits all three against the same WGAN/QuantGAN family on the same multi-asset universe is not currently standard practice. The forward diffusion process systematically corrupts real financial data $\mathbf{x}_0$ over T steps by adding Gaussian noise:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}) \tag{28}$$

where β_t dictates the variance schedule. A parameterized neural network θ then learns the reverse denoising process:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), {}_\theta(\mathbf{x}_t, t)) \tag{29}$$

Score-matching techniques allow models such as Fin-DDPM and CoFinDiff[47, 136] to seek improved distributional stability relative to selected GAN baselines, although comparisons remain largely within-paper. To impose domain-consistent dynamics, models such as GBM-Diff and Fin-Denoiser[137, 138] structurally incorporate geometric Brownian motion (GBM) priors[139] into diffusion stochastic differential equations (SDEs) to constrain baseline dynamics, although these priors do not by themselves guarantee financial validity.

Synthesis: macro-level market simulation and order flows

Macro-level simulation elevates the generative objective from simple price paths to the reproduction of complex market microstructures and multi-agent interactive environments. By generating realistic Limit Order Books (LOB) and order flows, these models facilitate the high-fidelity validation of execution strategies without exposing institutions to real capital risk.

Macro-level market simulation has progressed through three architectural generations, and a careful reading of the empirical evidence suggests that the strongest claims about the diffusion/LMM generation need to be qualified.

The first generation, conditional GANs for correlated price paths, was aimed at reproducing the cross-asset correlation structure that vanilla WGANs miss. Masi et al.[140] explicitly model correlated stock-market time series by injecting a learnable correlation block into the generator, evaluated

Table 5 | Comparative overview of financial time-series synthesis paradigms

Model Paradigm	Core methodology	Primary strengths	Key limitations
Wasserstein GANs	Adversarial minimax game with Earth Mover's distance	Rapid inference; highly effective for targeted metric alignment.	Susceptible to mode collapse; struggles with microstructural nuance.
Diffusion & SDEs	Iterative forward noising and reverse score-matching	Exceptional distributional fidelity; easily integrates financial priors (e.g., GBM).	Computationally intensive sampling process limits real-time utility.
Large Market Models	Transformer-based discrete sequence modeling	Scales to asynchronous tick data; enables interactive agent simulation.	Requires massive computational infrastructure and vast order-flow datasets.
DP-Generators	Differentially private gradient clipping	Ensures strict regulatory compliance and data anonymization.	Incurs a harsh trade-off between privacy bounds (ϵ) and data utility.

on the LOBSTER limit-order-book dataset. Istiaque et al.[141] extend this idea to *conditional* simulation, conditioning on macro state variables so that the generated regime matches a user-supplied label. Xia et al.[142] (Market-GAN) go further with a two-stage pipeline in which the first stage generates a market state and the second stage generates price paths conditioned on it. These three papers share a common limitation: their empirical evaluation is mostly on stylized-fact reproduction, and the downstream utility (i.e. "does the synthetic data actually improve trading agents trained on it?") is rarely reported. Kazadaev et al.[143] report a momentum-effect-specific benchmark on Moscow Stock Exchange data and show that GAN-synthetic trajectories reproduce the observed momentum premium; this is one of the few macro-simulation papers with an economic downstream task as the headline metric.

The second generation emerged from macroeconomic conditioning. Alex et al.[144] (MacroSynth) directly condition on macroeconomic factor panels (CPI, rates, credit spreads) so that generated market scenarios are internally consistent with a specified macro state. The reported evaluation is downstream mean-variance and risk-parity allocations computed on synthetic samples, with the synthetic allocations showing sign-consistent risk premia relative to history. The methodological advance here is that the conditioning signal is itself an economic primitive, not a latent regime inferred from the data—a useful design choice for stress-testing use cases where the user specifies the macro scenario in advance.

The third generation, diffusion and large market models, is the most recent wave of macro-simulation work and has moved to diffusion and discrete-token foundation models. Berti et al.[145] propose TRADES, a transformer-based denoising diffusion engine conditioned on market state for LOB order-flow simulation. The headline empirical claim is a 3.27- and 3.48-point improvement on the predictive score (MAE between real-data-trained and synthetic-data-trained price-prediction models) over the prior state of the art on two stocks. The authors themselves note that "there is a notable absence of quantitative metrics for evaluating generative market simulation models in the literature"—the field has been inventing its own evaluation protocols, which makes between-paper comparison fragile. Huang et al.[146] (DiGA / DigMA) split the problem into two stages: a conditional diffusion model over time-evolving distribution parameters of mid-price return rate and order arrival rate, and a meta-agent with financial economic priors that samples orders from the learned distributions. This decomposition sidesteps the difficulty of directly diffusing over discrete order sequences. Li et al.[147] (MarS) push the boundary furthest by training a Large Market Model (LMM)—a transformer foundation model over discrete order tokens, analogous to a language model over text—and report order-level interactive simulation that is responsive to user-injected orders.

Two important caveats qualify the evidence for this third generation of diffusion and large-market-model approaches. First, all three diffusion/LMM papers report improvements over a small set of in-paper baselines, not over a shared benchmark, so the apparent dominance of diffusion over GANs in macro simulation is a within-paper claim rather than a cross-paper empirical fact. Second, the diffusion and LMM papers report evaluation on at most two stocks (TRADES) or single datasets (DiGA, MarS); whether the reported fidelity survives a regime change (e.g. 2008 crisis, 2020 COVID) is not established. The survey's earlier claim that "macro-level LOB simulation and controllable text-to-market generation are still in their infancy" is therefore better supported than the implicit "diffusion vastly outperforms GANs" reading: the field has demonstrably advanced methodologically (transformer denoising, LMM-style discrete tokenization, meta-agent decomposition), but the empirical case for production deployment remains open. We flag this as a measurement gap the field would benefit from closing.

Synthesis: privacy-preserving and regulation-aware generation

The financial industry is heavily bottlenecked by strict data-sharing regulations (e.g., GDPR, CCPA). Synthetic data generation provides a mathematically viable pathway to share proprietary datasets across institutional boundaries without violating compliance standards. Models such as NVF-DPGAN[148] directly incorporate differential privacy (DP) mechanisms[149,150] into the generative training process. By clipping gradients and injecting calibrated noise during optimization, the framework provides a formal (ϵ, δ)-DP bound on membership inference[151]; this bound limits, but does not eliminate, inference about individual records or proprietary signals. The ongoing central research focus in this sub-domain is optimizing the delicate trade-off between strict privacy bounds (ϵ) and the downstream predictive utility of the resulting synthetic series.

Synthesis: emerging trends

Recent literature reflects a rapid maturation of financial synthesis, particularly transitioning away from unconditional generation toward highly controllable, context-aware frameworks. Rather than treating synthesis as a purely statistical exercise, recent models simultaneously condition on diverse metadata and partially observed signals to dictate the semantic direction of the synthetic output[152].

For instance, Lozano et al.[41] propose advanced dual-sourced diffusion networks that align text-based market sentiment directly with numerical trajectory generation, ensuring exceptionally high temporal coherence across modalities. Parallel to improvements in fidelity, there is a critical push to reduce the computational burden of generative sampling. Tao et al.[153] introduce distillation techniques[74] specifically designed to accelerate the generation process, a critical requirement for deploying synthetic simulators in live, high-frequency trading environments. Finally, Oppel et al.[154] demonstrate how these advanced conditional generators are practically utilized, embedding them into dynamic stress-testing pipelines to evaluate institutional portfolio resilience under highly specific, synthetically engineered counterfactual crises.

Synthesis: research synthesis

Data synthesis currently represents the most dynamic and rapidly evolving branch of Financial Time-Series Generation (FTSG). As summarized in Table 5, the field relies on a diverse array of generative architectures, each serving distinct institutional needs. While micro-level return generation has become comparatively well developed-reliably capturing volatility clustering and cross-asset correlations via diffusion models-macro-level LOB simulation and controllable text-to-market generation are still in their infancy. Moving forward, the transition towards computationally efficient, privacy-preserving, and condition-specific market simulators can lower

barriers to algorithmic trading research and enhance the capability of financial institutions to conduct rigorous, counterfactual stress testing.

Applications: extrapolation and the application landscape

This section zooms out to the downstream use-cases that actually justify an FTSG model in a quantitative-finance workflow. We organize the discussion along the task axis (FTSE, FTSI, FTSS) and end with a cross-cutting paragraph on shared infrastructure (RL training gyms, risk-management sandboxes, and synthetic-data sharing). Throughout, we restrict ourselves to peer-reviewed or peer-reproducible works whose authors explicitly evaluate against a real downstream application rather than only on statistical fidelity.

FTSE is the most mature sub-task in deployment. The classical use-case is to feed extrapolated point or quantile forecasts into a downstream decision module, and the strongest evidence that this works comes from the deep-reinforcement-learning (DRL) literature. FinRL[155] and its successor FinRL-Meta[156] treat pretrained time-series forecasters (LSTM, Transformer, or more recently LLM-based adapters) as the observation pipeline for PPO/A2C/SAC trading agents[157–159] trained on Dow-30 and S&P-500 constituents; their empirical study reports that agents whose observation encoders are first fine-tuned on FTSE targets achieve higher Sharpe ratios[75] and lower maximum drawdown than agents trained on raw OHLCV. Hierarchical extensions such as HRPM[160] couple a portfolio-management FTSE head with a separate order-execution head, explicitly accounting for price slippage when large extrapolation errors are amplified by capital allocation. Time-LLM[19] and CALF[67] are representative financial forecasters with adapter architectures related to these workflows; direct, standardized FinRL comparisons remain limited.

A second, often-overlooked use of FTSE is volatility and risk forecasting. TAIL-GAN[133] is explicitly designed to satisfy the joint elicitability of Value-at-Risk and Expected Shortfall, and its authors plug the tail-respecting generator into a VaR back-test to examine coverage under VaR/ES back-testing criteria[135]. On the option-pricing side, deep generative models of the underlying (e.g., quant GAN variants) have been used to produce Monte-Carlo paths[161] that replace historical resampling for pricing exotic payoffs, where the FTSE tail behavior dominates pricing error. FTSE outputs also feed stochastic-portfolio-optimization engines: Hybrid CGAN for Portfolio Analysis[162] trains a conditional GAN whose label is the Markowitz risk-aversion level[163], and the resulting synthetic future paths are scored by an optimizer that selects weights without committing to a single forecast.

Applications: imputation

FTSI is most often deployed in production at the data-engineering layer, where its outputs feed classical pipelines rather than a learning agent. Yoon et al.[10] introduced GAIN, the generative-adversarial imputation paradigm, and it remains a baseline in many open finance datasets for handling missing quotes during off-hours, ticker delistings, and exchange halts. Subsequent work has tightened the fit between GAN imputation and financial time structure: ImputeGAN[101] and AttnWGAIN[100] replace generic CNN backbones with Informer and diagonal-masked attention respectively, both reporting smaller distributional drift on the K-line dataset than linear interpolation or MICE baselines[86]. Beyond price reconstruction, missing-data scenarios in finance also arise at the macro layer. FCI[164] uses Bayesian factor imputation to keep Financial Condition Indices computable during incomplete-reporting periods (e.g., during COVID-era reporting lags), demonstrating that imputed macro factors can materially change the sign of the indicator relative to linear interpolation.

A second family of applications concerns downstream classifier performance on financial tabular data. For instance, Emaan[165] reports that Transformer-enhanced GAN oversampling for synthesizing minority fraud transactions yields substantial gains in detection recall compared to heuristic oversampling methods such as SMOTE[166] on standard credit-card transaction benchmarks; similarly, in credit-scoring ensembles[167], GAN-augmented minority classes materially reduce false-negative rates. These studies reinforce that, in the financial setting, generative imputation and augmentation are rarely the final product; rather, they serve as value-adding pre-processing steps whose utility is judged by downstream predictive metrics.

Applications: synthesis

FTSS is the sub-task with the broadest and most consequential deployment surface. We organize it along the natural lifecycle of a quantitative strategy.

ABIDES[168] and its Gym-compatible successor ABIDES-Gym[169] are agent-based discrete-event simulators that wrap a synthetic limit-order-book generator in an OpenAI Gym interface[170], enabling head-to-head backtests of RL agents that interact with thousands of background traders. Balch et al.[171] compare single-agent market replay against multi-agent interactive simulation and find that the interactive mode systematically reorders the leaderboard of public trading algorithms — a striking demonstration that the choice of an FTSS simulation backbone can flip strategy rankings. Rollins and Cliff[172] extend this with a threaded parallel simulator, scaling ABIDES-style order generation to high agent counts, and reproduce qualitatively different "pecking orders" of strategies compared with single-agent replay.

GAN- and diffusion-based synthesizers are increasingly used to manufacture counterfactual tail scenarios that historical samples cannot supply. TAIL-GAN[133] produces tail scenarios that satisfy joint VaR/ES elicitability and are consumed by bank-level stress-testing frameworks; the authors explicitly demonstrate that the synthesized tails perform well in VaR/ES back-tests where naive block bootstrap[173] performs less reliably. Hybrid CGAN for Portfolio Analysis[162] generates market scenarios conditioned on risk aversion, allowing a portfolio engine grounded in mean–variance optimization[163] to be re-evaluated against thousands of hypothetical regimes in seconds. On the macro-finance side, MacroSynth[144] conditions on macroeconomic factors to produce co-occurring crash and bull-run scenarios, and the authors show that downstream mean–variance and risk-parity allocations[174] computed on the synthetic samples exhibit sign-consistent risk premia relative to the historical record. Diffusion-based generators are entering the same niche: CoFinDiff[47] is evaluated as a scenario engine for a portfolio whose weights are re-optimized on each generated window, and Fin-Denoiser[138] supplies high-SNR scenarios for risk models that are sensitive to data artifacts.

Perhaps the most visible deployment of FTSS is as a training environment for RL trading agents. FinRL-Meta[156] ships near-real market environments built from Yahoo Finance and CCXT feeds and demonstrates that DRL agents pre-trained on these synthetic-augmented environments generalize better to the held-out test year than agents trained only on the historical window — a result that is reproduced across DOW-30, S&P-500 and crypto-asset universes. MarS[147] extends this idea to the order level by generating asynchronous event streams, allowing agents to be evaluated on metrics such as realized spread and adverse-selection cost[175, 176] that are invisible to bar-level environments. Robust Risk-Sensitive RL Agents[177] stress-test the resulting agents by perturbing the synthetic environment, showing that risk-aware training reduces performance variance by an order of magnitude across random seeds. Collectively, these works establish FTSS as the substrate for "sim-to-real" transfer in financial RL.

In privacy-preserving data sharing and regulatory scenarios, synthetic financial data is a recognized route to overcome the friction between data sharing and the privacy constraints imposed by GDPR, CCPA, and the equivalent Chinese PIPL. Balch et al.[178] propose the Six Levels of Privacy framework and demonstrate empirically that synthetic equities, options and LOB samples generated by TimeGAN, QuantGAN, and diffusion variants preserve aggregate utility (Sharpe ratios, correlation matrices, tail-event counts) while reducing vulnerability to membership-inference attacks[151] to near-baseline. Boudewijn et al.[179] formalize the corresponding evaluation protocol that quantifies the privacy/utility frontier. On the regulatory side, Balch et al.[178] further argue that FTSS pipelines can produce synthetic benchmark datasets for third-party model validation without exposing proprietary positions, a direction that has begun to attract policy and

supervisory attention in public-sector pilot studies. This shift reframes FTSG not only as a research artifact but as compliance infrastructure.

Cross-cutting applications: synthetic datasets as benchmarks

Beyond the per-task applications above, FTSG has produced a small but rapidly maturing ecosystem of benchmark datasets that are themselves reusable scientific infrastructure. TSGBench[180] provides paired real/synthetic time-series collections for GOOGL and FX series and is consumed by downstream papers as a standardized fidelity benchmark. CTBench[181] extends this to Binance cryptocurrency markets, where extreme volatility makes synthetic-data quality both harder to achieve and more consequential. FinTSB[27] and FinTSBridge[182] package millions of Chinese A-share, NYSE and NASDAQ trajectories together with pre-defined train/valid/test splits, lowering the cost of entry for new FTSG research and making FTSE/FTSI results directly comparable. These benchmarks invert the typical deployment pipeline: rather than synthesizing data to fit a model, the model is now fitted to a benchmark whose provenance, statistics and license are publicly auditable. This is a subtle but important shift for reproducibility and regulatory acceptance.

In summary, FTSE, FTSI, and FTSS are not merely three academic formulations of the same underlying problem. Each maps onto a distinct, value-generating workflow in industry. FTSE plugs into decision engines (trading agents, portfolio optimizers, risk models); FTSI into data-engineering layers (data cleaning, augmentation for tabular classifiers); and FTSS into simulation infrastructure (RL training, stress testing, privacy-preserving sharing). The applications in this section collectively motivate the breadth of design choices catalogued in the taxonomy and task-specific generation subsections.

Evaluation: why generic generation metrics are insufficient

Evaluation remains one of the most critical, yet under-standardized, aspects of financial time-series generation. In conventional supervised learning or computer vision tasks, performance is often adequately measured by generic metrics like Mean Squared Error (MSE), Fréchet Inception Distance (FID)[183], or visual realism. However, FTSG involves multiple competing objectives where visual similarity or pointwise accuracy does not equate to financial validity.

A synthetic sequence might achieve a low MSE or MAE against a ground-truth trajectory, yet entirely fail to preserve crucial dynamics such as volatility clustering or tail-risk behavior, limiting its usefulness for stress testing. Conversely, generic metrics cannot evaluate whether generated data improves a trading agent's performance in a downstream simulation. Financial data's noise, non-stationarity, and regime-dependent nature demand that evaluation protocols go beyond generic reconstruction. In this section, we systematically organize evaluation practices into five dimensions, culminating in a unified evaluation checklist.

Evaluation: fidelity-oriented assessment

A primary objective of Financial Time-Series Generation (FTSG) is to approximate the underlying multivariate data-generating distribution closely. Fidelity-oriented evaluation rigorously quantifies the statistical divergence between the synthetic sequences $\hat{\mathbf{X}}$ and the empirical observations $\mathbf{X}$. This assessment spans several mathematical dimensions:

Distributional similarity evaluates whether the model captures the marginal distributions of returns or prices through optimal transport and probabilistic divergences. The 1-Wasserstein distance[34, 184] between the empirical distribution $\mathbb{P}_r$ and the generated distribution $\mathbb{P}_g$ measures the minimum cost required to transform one distribution into the other:

$$W_1(\mathbb{P}_r, \mathbb{P}_g) = \inf_{\gamma \in (\mathbb{P}_r, \mathbb{P}_g)} \mathbb{E}_{(x,y)\sim\gamma}[\| x - y \|] \tag{30}$$

where $(\mathbb{P}_r, \mathbb{P}_g)$ is the set of all joint distributions with marginals $\mathbb{P}_r$ and $\mathbb{P}_g$. Additionally, the maximum mean discrepancy (MMD)[185] evaluates distances in a reproducing kernel Hilbert space (RKHS)[91] $\mathcal{H}$ with kernel k:

$$\mathrm{MMD}^2(\mathbb{P}_r, \mathbb{P}_g) = \mathbb{E}[k(x, x')] + \mathbb{E}[k(y, y')] - 2\mathbb{E}[k(x, y)] \tag{31}$$

where $x, x' \sim \mathbb{P}_r$ and $y, y' \sim \mathbb{P}_g$.

For temporal dependence, financial data are fundamentally sequential. The generator must capture auto-regressive dynamics and lead-lag relationships. This is mathematically verified by comparing the Autocorrelation Function (ACF) of the generated returns $\hat{r}_t$ with the empirical returns r_t. For a given lag k, the ACF is defined as:

$$\rho_k = \frac{\mathrm{Cov}(\mathrm{r_t}, \mathrm{r_{t-k}})}{\mathrm{Var}\,(\mathrm{r_t})} \tag{32}$$

The Mean Absolute Error (MAE) between the empirical ACF vector and the synthetic ACF vector serves as a standard metric for temporal fidelity.

For cross-sectional dependence, the covariance structure among assets in multivariate generation dictates portfolio-level risk. Cross-sectional fidelity is evaluated by computing the Frobenius norm of the difference between the empirical correlation matrix Σ_r and the synthetic correlation matrix Σ_g:

$$\mathcal{E}_{corr} = \| \Sigma_r - \Sigma_g \|_F = \sqrt{\sum_{i=1}^{N}\sum_{j=1}^{N} (\sigma_{r,ij} - \sigma_{g,ij})^2} \tag{33}$$

Spectral and frequency-domain similarity checks that cyclical macroeconomic patterns and the low signal-to-noise ratio are maintained by applying the fast Fourier transform (FFT)[186] to convert sequences into the frequency domain. The Power Spectral Density (PSD) of the synthetic data is then compared against the real data to penalize generators that produce over-smoothed, frequency-deficient signals.

Evaluation: financial validity assessment

A distinctive requirement of FTSG is that generated sequences must adhere to established financial theories and principles. High distributional similarity does not guarantee the preservation of empirical "stylized facts"-recurrent non-linear statistical regularities observed across many financial markets[14]. Evaluation should therefore test the presence of these facts:

Heavy tails and kurtosis are assessed because financial returns are commonly non-Gaussian, exhibiting excess kurtosis and fat tails. To quantify the generator's ability to produce extreme tail events (e.g., black swan crashes), researchers measure the sample kurtosis $\kappa = \frac{\mathbb{E}[(r_t-\mu)^4]}{\sigma^4}$ and estimate the tail index α using the Hill estimator[187]. For the upper tail, ordering observations as $X_{(1)} \geq X_{(2)} \geq \cdots \geq X_{(n)}$, the Hill estimator for a threshold k is:

$$\hat{\alpha} = \left(\frac{1}{k} \sum_{i=1}^{k} \ln X_{(i)} - \ln X_{(k+1)} \right)^{-1} \tag{34}$$

The synthetic sequence must yield an $\hat{\alpha}$ closely matching that of historical data.

Volatility clustering, documented by Mandelbrot[126] and systematized among stylized facts by Cont[14], is assessed as a market property in which large price variations are followed by large variations. A valid generator must replicate this slow-decaying memory in variance. This is evaluated by computing the autocorrelation of squared returns r_t^2 or absolute returns $|r_t|$ over extended lags, verifying that $\rho_k(|r_t|) > 0$ for large k.

The leverage effect[14, 188] is assessed as the asymmetric phenomenon in which negative returns increase future volatility more severely than positive returns of the same magnitude. It is rigorously quantified by the cross-

Table 6 | A unified evaluation checklist for financial time-series generation

Evaluation Dimension	Key Metrics / Methods	Primary task suitability	Reflects financial validity?
Point-wise Accuracy	MSE, MAE, RMSE, MAPE	Extrapolation, Imputation	Low (Ignores market dynamics)
Distributional Fidelity	Wasserstein Distance, MMD, KL Divergence, Cross-Correlation	Synthesis, Extrapolation	Medium (Captures static shapes)
Stylized Facts	ACF of squared returns, Tail Index, Kurtosis, Leverage effect tests	Synthesis, Simulation	High (Domain-specific realism)
Downstream Utility	Train-on-Synthetic, Test-on-Real (TSTR); Train-on-Real, Test-on-Real (TRTR); Sharpe Ratio of trained agents; VaR/ES accuracy	All Tasks	High (Practical applicability)
Privacy & Security	Distance-to-Closest Record (DCR), Differential Privacy ϵ	Synthesis (Data Sharing)	N/A (Focuses on compliance)

correlation between past returns and future squared returns at lag $\tau > 0$:

$$L(\tau) = \mathrm{Corr}\,(\mathrm{r_t}, \mathrm{r}^2_{\mathrm{t}+\tau}) \tag{35}$$

A financially valid synthetic dataset must exhibit $L(\tau) < 0$ for small τ.

Tail risk preservation, assessed through VaR and ES, is checked because synthetic data used for institutional stress-testing must not systematically underestimate market risk. Generative fidelity is audited using Value-at-Risk (VaR) and Expected Shortfall (ES)[134,135]. For a confidence level $1-\alpha$ (e.g., 99%), VaR is the quantile of the return distribution:

$$\mathrm{VaR}_\alpha = \inf\{z \in \mathbb{R} | \mathbb{P}(r_t \le z) \ge 1-\alpha\} \tag{36}$$

While ES measures the expected loss conditional on exceeding the VaR threshold:

$$\mathrm{ES}_\alpha = \mathbb{E}[r_t | r_t \le \mathrm{VaR}_\alpha] \tag{37}$$

Significant deviations between empirical and synthetic VaR/ES indicate a dangerous smoothing of downside risk.

Evaluation: downstream utility assessment

To bridge the gap between theoretical machine learning metrics and practical finance, an increasingly adopted paradigm evaluates generated data primarily by utility in executing downstream financial tasks. Improved downstream performance provides task-specific evidence of synthetic-data value.

Forecasting improvement, assessed through TSTR versus TRTR, evaluates the generator as a data augmentation engine. A predictive model f_θ is trained entirely on generated data $\hat{\mathbf{X}}$ and evaluated on real, out-of-sample data $\mathbf{X}_{test}$. This train-on-synthetic/test-on-real (TSTR) performance, following the evaluation design of Esteban et al.[189], is compared against the train-on-real/test-on-real (TRTR) baseline. A TSTR predictive error (e.g., MSE or directional accuracy) that closely approaches or surpasses TRTR demonstrates exceptional generative utility.

Trading-agent training and simulation is evaluated by deploying the generative model as an interactive market environment in which reinforcement learning (RL) trading agents can be trained on millions of synthetic episodes. The utility of the generator is subsequently quantified by the financial performance of the agent when deployed on real out-of-sample data, measured via the annualized Sharpe ratio (SR)[75] and maximum drawdown (MDD):

$$SR = \frac{\mathbb{E}[R_p - R_f]}{\sqrt{\mathrm{Var}\,(R_p)}}, \quad MDD = \max_{\tau \in (0,T)} \left(\frac{\max_{t \in (0,\tau)} P_t - P_\tau}{\max_{t \in (0,\tau)} P_t} \right) \tag{38}$$

where R_p is portfolio return, R_f is the risk-free rate, and P_t is the portfolio value at time t.

Evaluation: robustness, controllability, and privacy

As generative architectures-particularly foundation models and diffusion models-become increasingly sophisticated, evaluation protocols must expand to audit advanced capabilities and regulatory compliance.

Regime robustness is assessed because generic metrics calculated over long, blended time horizons obscure a model's failure in specific contexts. Robustness evaluation partitions test data into discrete regimes (e.g., low-volatility bull markets vs. high-volatility crashes) using hidden Markov models (HMMs)[129,130] or variance thresholds. Models are evaluated on their ability to maintain stable generative fidelity across these shifting boundaries.

Conditional consistency is assessed for multimodal or controllable generation (e.g., text-to-market or target-volatility synthesis). Evaluation must quantify semantic alignment. This involves measuring the correlation between the conditioning input (e.g., a target VaR level) and the actualized metric in the generated sequence, ensuring the generator is not merely ignoring the prompt to produce unconditional mode-collapsed outputs.

Privacy leakage and memorization risk are assessed because a critical motivation for FTSG is reducing privacy risk under financial data-sharing regulations (e.g., GDPR). To verify that a model has not simply memorized proprietary training data, the distance-to-closest-record (DCR) statistic, used in privacy-preserving synthetic-data evaluation[190], is computed:

$$\mathrm{DCR}\,(\hat{x}) = \min_{x \in \mathbf{X}_{train}} \| \hat{x} - x \|_2 \tag{39}$$

If the DCR distribution peaks near zero, the model suffers from severe privacy leakage. Furthermore, models optimized with differential privacy (DP)[149,150] are evaluated against their (ϵ, δ)-DP bounds, analyzing the mathematical trade-off between the privacy budget ϵ and the resulting degradation in downstream utility.

Evaluation: a unified checklist

The lack of standardized evaluation frequently leads to contradictory claims regarding state-of-the-art performance in FTSG. To resolve this, we formalize a unified evaluation checklist, mapping the aforementioned rigorous metrics to their appropriate problem formulations.

As detailed in Table 6, researchers must construct a composite evaluation protocol. For instance, an *imputation* model may prioritize point-wise accuracy (MSE) and temporal coherence, whereas a *market simulation* architecture must be rigorously audited against stylized facts (kurtosis and Ljung–Box tests[191] for temporal dependence) and downstream utility (TSTR). No single metric is sufficient; financial validity emerges only at the intersection of distributional fidelity, economic logic, and practical applicability.

Evaluation: emerging trends and benchmarking

The evaluation of synthetic financial data is rapidly transitioning from the isolated calculation of generic metrics toward comprehensive, standardized benchmark platforms. Recent literature indicates a shift in focus on multimodal reasoning, structural market constraints, and full-pipeline financial

workflows. We categorize these emerging trends into three primary directions:

The first direction is conditional specificity and scenario-based evaluation: standard distribution-matching metrics often fail to evaluate whether a conditional generative model actually respects its conditioning metadata (e.g., specific market regimes or macroeconomic textual prompts). To address this, Narasimhan and Jiang[39] introduced *TimeWeaver*, together with specificity metrics that explicitly penalize generative models for failing to reproduce metadata-specific features in the synthesized sequences.

The second direction is market microstructure and LOB standardization: evaluating high-frequency tick data and Limit Order Books (LOB) requires specialized protocols due to their asynchronous nature and stringent cross-feature constraints. Zhong et al.[192] recently introduced *LOBench*, a standardized benchmark specifically engineered for LOB representation and generation. Utilizing massive, real-world China A-share market data, LOBench establishes unified preprocessing pipelines and task-specific baselines, allowing researchers to evaluate generative fidelity directly against microstructural properties such as queue dynamics and volume imbalances.

The third direction is full-pipeline workflows and multimodal reasoning: instead of treating time-series generation as an isolated algorithmic task, modern benchmarks evaluate synthetic data based on its utility within broader economic workflows. Jiang et al.[193] proposed *FinMaster*, a holistic benchmark featuring a dedicated *FinSim* module that evaluates how well generated synthetic, privacy-compliant financial data replicates market dynamics for downstream corporate auditing and accounting tasks. Similarly, Yu et al.[194] introduced the multi-modal benchmark *TSRBench* to stress-test the reasoning capabilities of Large Language Models (LLMs) over time-series data. These platforms reveal a critical insight for evaluation: strong semantic reasoning does not guarantee accurate context-aware numerical generation, highlighting the need for hybrid evaluation metrics that simultaneously score textual alignment and continuous mathematical fidelity.

Data resources: raw sources and modalities

The empirical foundation of Financial Time-Series Generation (FTSG) relies on the quality, granularity, and structure of the underlying data. Unlike static image datasets, financial data is continuous, highly non-stationary, and prone to hidden biases. We systematically cluster these resources moving from raw modalities to curated databases, standardized benchmarks, and ultimately, the practical pitfalls of dataset construction.

Financial time series are not monolithic; they exhibit drastically different statistical signatures depending on the asset class and sampling frequency. The rationale for clustering these sources lies in the specific generative challenges they present. Raw FTSG data fall into four distinct modalities.

Equities and market indices encompass individual stocks (e.g., AAPL, TSLA) and aggregate indices (e.g., S&P 500, CSI 300). Sampled typically at daily or minutely frequencies, these series are the primary testbeds for unconditional synthesis and extrapolation. They present fundamental generative challenges, including pronounced *volatility clustering* and *fat-tailed distributions*[14], as well as *regime-shifting non-stationarity*[130].

High-frequency microstructure data include limit order books (LOB), tick-by-tick trades, and Level-2 quotes recorded at the millisecond or microsecond level. The market-microstructure properties of LOB data[76,77] make generation complex due to its asynchronous nature and the strict spatial constraints required (e.g., bids must strictly remain below asks, and queue volumes dictate execution). This data is essential for simulating interactive, multi-agent algorithmic trading environments.

Macroeconomic and fixed-income data include indicators such as Treasury yield curves, the Consumer Price Index (CPI), and Gross Domestic Product (GDP). Sampled at low frequencies (monthly or quarterly), these datasets reflect systemic economic health. The primary challenge here is severe data sparsity and the need to interpolate/impute missing values across shifting long-term economic cycles.

Alternative data and textual metadata augment purely numerical series as multi-modal foundation models become more common. This includes macroeconomic news headlines, central bank policy texts, earnings call transcripts, and social media sentiment[195,196]. This modality is crucial for training controllable generators and scenario-guided market simulators.

Data resources: curated databases and APIs

Historically, raw financial data is fragmented and optimized for human visualization. To facilitate the massive data ingestion required by deep generative models, researchers routinely aggregate data from institutional databases and developer APIs.

For rigorous backtesting and synthesis, institutional and academic repositories including Wharton Research Data Services (WRDS), the Center for Research in Security Prices (CRSP), and Compustat[197–199] are designed to mitigate the survivorship bias documented by Brown et al.[200]. For macroeconomic series, FRED (Federal Reserve Economic Data)[201] is a widely used public repository maintained by the Federal Reserve Bank of St. Louis. In Chinese markets, CSMAR[202] offers comprehensive, high-quality A-share trading records.

Open-source and developer APIs are also available through platforms such as Yahoo Finance and Investing.com[203,204], frequently accessed via the open-source Python package `yfinance`[205]. For programmatic, high-frequency access, APIs such as Alpha Vantage, Alpaca, and Polygon.io[206–208] deliver massive historical streams suitable for training large-scale predictive models.

Cryptocurrency exchanges are widely used because 24/7 trading and open data access make crypto markets particularly suitable for testing generative algorithms. Platforms such as Binance and CoinMarketCap[209,210] provide free API access to granular tick and volume data, capturing extreme volatility events ideal for stress-testing generation.

Data resources: standardized benchmarks and evaluation suites

Historically, FTSG research relied on ad-hoc, proprietary data slices, severely hindering reproducibility and fair model comparison. The recent emergence of unified benchmark platforms marks a critical maturation in the field. Recent benchmark studies, including FinTSB[27], GIFT-Eval[211], TFB[212], FinTSBridge[182], TSGBench[180], and CTBench[181], encapsulate vast datasets, pre-define chronological train/test/validation splits, and standardize evaluation metrics (e.g., masking ratios for imputation, prompt templates for forecasting).

By providing a "level playing field," these suites allow researchers to disentangle algorithmic improvements from data-mining advantages. A summary of recent and authoritative FTSG benchmarks is provided in Table 7andTable 8.

Data resources: practical issues in dataset construction

Constructing financial datasets for generative models is fraught with domain-specific pitfalls. A generative architecture, no matter how mathematically advanced, will learn and amplify the structural biases present in its training data. Five issues are particularly important.

Survivorship bias[200] occurs when datasets that only include currently active assets inadvertently filter out companies that went bankrupt or were delisted. Generative models trained on such data will synthesize artificially optimistic market trajectories and fail to accurately model the probability of asset default, ruining their utility for downside risk management.

Look-ahead bias in alignment occurs when models are trained on information that was not publicly available at the timestamp of generation. A common pitfall in FTSG is using historically *revised* macroeconomic data (e.g., revised GDP figures published months later) rather than the point-in-time, unrevised data emphasized in leakage-aware financial machine learning[213], causing the generator to leak future realities.

Chronological splitting, rather than random masking, is required because unlike static images where independent and identically distributed (i.i.d.) random splitting is acceptable, financial time series must generally be partitioned chronologically, using temporal validation or purged and

Table 7 | Summary of recent financial time-series generation benchmarks

Dataset / Benchmark	Tasks	Primary data sources	Split ratio	Scale (Approx.)
FinTSB[27]	Extrapolation / Imputation	Chinese A-share, NYSE, NASDAQ	7:1:2	150M points
GIFT-EVAL[221]	Forecasting	27 datasets (Stock, Crypto, FX, etc.)	Various	144,380 series
TFB[28]	Forecasting	Broad finance domain	Various	2.1M series
FinTSBridge[182]	Extrapolation / Imputation	S&P 500, NASDAQ, FTSE, CSI 300	7:1:2	86,978 series
TSGBench[180]	Synthesis / Simulation	GOOGL, Global Exchanges	9:0:1	13,213 series
CTBench[181]	Synthesis	Cryptocurrencies on Binance	9:0:1	530 series
FinMultiTime[222]	Extrapolation	S&P 500, HS 300	N/A	6M points

Table 8 | Literature retrieval and screening statistics

Screening stage	Count	Retained	Primary reason for exclusion
Initial Keyword Search	2568	2568	N/A
Title/Abstract Screening	2568	604	Not financial data / Pure deterministic forecasting
Full-text Review	604	**131**	Weak financial validation / Duplicated preprints
Primary task assignment			
7D2Extrapolation		54	
7D2Imputation		29	
7D2Synthesis / Simulation		47	

Each of the 131 retained studies is assigned to exactly one primary task category; cross-cutting contributions (e.g., forecasting with imputation or synthesis with scenario conditioning) are discussed in the relevant sections but are not double-counted.

embargoed splits where appropriate[213]. Randomly splitting overlapping sequence windows leads to severe temporal data leakage. Similarly, in imputation tasks, random masking often fails to reflect the missingness mechanisms distinguished by Rubin[80] (e.g., contiguous trading halts), leading to models that perform well on paper but fail in live environments.

Asynchronous frequencies and missingness arise in multivariate generation (e.g., simulation of a global portfolio) because assets trade on different global exchanges with distinct operating hours and holidays, creating non-synchronous observations[81]. Naively padding these datasets forward creates artificial zero-volatility spans, causing cross-sectional generators (such as diffusion models) to learn spurious flat-line correlations.

Market regime imbalance, consistent with regime-switching models[130], occurs because historical datasets are overwhelmingly dominated by standard, low-to-medium volatility market conditions. Without specialized sampling techniques, models suffer from severe class imbalance and will systematically fail to generate realistic crash scenarios (e.g., the COVID-19 market reaction in 2020[214]) simply because these events represent a fraction of the training distribution.

Discussion

Despite the rapid advancements in financial time-series generation, several fundamental challenges remain unresolved at the intersection of generative artificial intelligence and empirical financial econometrics. Bridging the gap between statistical expressiveness and market validity requires addressing core bottlenecks in causality, theoretical boundedness, cross-modal integration, and temporal non-stationarity.

A first fundamental challenge concerns causal and invariant representation learning: current FTSG models often learn spurious correlations specific to a particular market environment or training window. This stems from a lack of explicit disentanglement between true causal factors (e.g., shifts in fundamental central bank policies) and mere coincidental patterns (e.g., a specific high-frequency trading algorithm's temporary market dominance). Consequently, models that assume data are Independent and Identically Distributed (i.i.d.) can degrade substantially when encountering novel market regimes due to severe out-of-distribution (OOD) shifts[57, 215]. To address this, future FTSG frameworks must move toward environment-invariant learning. By treating specific market regimes (e.g., high vs. low volatility) as distinct training "environments," techniques like Invariant Risk Minimization (IRM)[57] can be used to reduce sensitivity to spurious environmental correlations. Early attempts in forecasting demonstrate that enforcing representation invariance across regimes improves robustness[58]. Porting this paradigm to generative models may improve the robustness of synthesized trajectories under novel economic shifts.

A second challenge is econometrically bounded generation. A well-documented constraint of purely data-driven generative models (such as standard GANs[15] and diffusion models[16, 37]) is that they can replicate past empirical distributions without guaranteeing adherence to established market microstructure or stochastic process theory[14]. For instance, unconstrained models might generate prices that breach arbitrage boundaries or fail to exhibit mean-reverting volatility. Drawing inspiration from Physics-Informed Neural Networks (PINNs) in the physical sciences[216], future research should focus on *econometrically-bounded* generation. Embedding economic priors—such as geometric Brownian motion (GBM)[139], stochastic volatility models, or jump-diffusion processes for extreme events[217]—directly into the generator's loss functions provides a powerful white-box constraint. Recent advancements have successfully integrated GBM priors into score-matching diffusion models[47]. Expanding this to capture more complex stylized facts will ensure that synthesized sequences are not merely statistically plausible, but also economically rational.

A third challenge is multi-modal time series generation. Existing FTSG models broadly generate time-series data in isolation, neglecting the exogenous, multi-modal signals that drive actual market dynamics. However, financial markets are multi-causal systems: a macroeconomic announcement (textual metadata) triggers shifts in order flow (discrete events), which subsequently alter implied volatility surfaces (continuous manifolds). To capture this complexity, future work should pursue two promising avenues. First, cross-modal conditioned generation can encode broader heterogeneous data, such as central bank operations, limit order book states, and earnings call transcripts [40, 41, 152], using a unified representation space to precisely steer generative trajectories beyond narrow textual prompts [67, 218]. Second, joint multi-modal generation involves the simultaneous generation

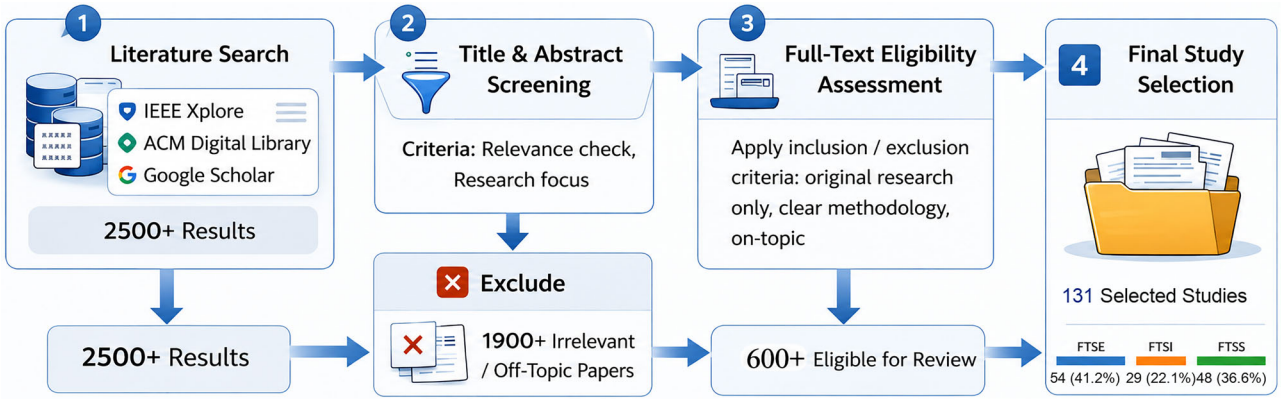


Fig. 6 | Literature identification and PRISMA screening pipeline. Flow diagram illustrating the systematic retrieval, multi-stage screening, and eligibility determination process compliant with the PRISMA 2020 framework. Blue rectangular boxes represent initial literature identification across primary databases (Google Scholar, DBLP, Web of Science, Scopus, arXiv) and duplicate removal. Orange rectangular boxes denote title and abstract screening stages with corresponding exclusion criteria. Red dashed borders and rightward branching callouts indicate excluded records at each phase along with explicit rationales (non-financial domains, deterministic point forecasting without generative distributions, non-reproducible records). Green solid rectangular boxes indicate the final corpus of 131 retained studies meeting all eligibility criteria, partitioned into three task categories: financial time-series extrapolation, imputation, and data synthesis. Detailed quantitative counts for each filtering phase are recorded in the accompanying retrieval statistics table.

of linked data streams. By jointly modeling limit order books, financial news headlines, and derivative prices, generators can inherently learn cross-modal causal dependencies and lead-lag relationships [41], which is vital for simulating high-impact scenarios such as flash crashes.

A fourth challenge is online adaptation under regime shifts. Despite the zero-shot capabilities of modern time-series foundation models, they remain susceptible to concept drift—the frequent distributional shifts triggered by evolving economic cycles or regulatory changes. Therefore, static generative models will inevitably degrade over time. To avoid the notorious "catastrophic forgetting" associated with fine-tuning on new data [78,79], FTSG must transition toward online adaptation. One promising avenue is Bayesian Continuous Learning, utilizing Streaming Variational Bayes [219] to incrementally update the generator's latent space as new market observations arrive in real time. Alternatively, framing generation within an Online Reinforcement Learning framework [220], where the model receives rewards based on the real-time fidelity of its synthetic samples relative to current market conditions, allows for near-instant calibration to abrupt regime shifts without retraining from scratch.

Finally, we reflect on limitations, practical implications, and a concluding outlook. While this survey synthesizes 131 studies spanning 2021 to early 2026, several intrinsic limitations characterize the current empirical literature. Primarily, existing benchmarks remain disproportionately concentrated in liquid equity and cryptocurrency markets, with a comparative scarcity of standardized suites for fixed income, credit derivatives, and illiquid alternative assets. Furthermore, the absence of universally accepted, open-source matching engine simulators hampers standardized, apples-to-apples comparisons of downstream execution utility. Nonetheless, our proposed two-level taxonomy successfully unifies the field across task-oriented and technique-driven dimensions, providing a structured conceptual framework for future studies. By compiling curated databases, establishing a multi-dimensional evaluation protocol, and identifying theoretical boundaries, this review provides a structured foundation for researchers and practitioners developing next-generation financial generative models.

Methods

This review follows the reporting logic of PRISMA 2020 for transparent search, screening, eligibility assessment, and synthesis[29]. To ensure the reproducibility, transparency, and comprehensiveness of this review, we adhered to a systematic literature review methodology tailored to the intersection of artificial intelligence and quantitative finance, designed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) guidelines[29]. The completed PRISMA 2020 checklist is provided as Supplementary Information. This section outlines our protocols for literature retrieval, inclusion/exclusion criteria, data extraction, and the dimensions utilized for synthesizing the gathered research.

Literature retrieval strategy

Our primary objective was to capture the state-of-the-art advancements in financial time-series generation, particularly focusing on the paradigm shift driven by generative AI over recent years.

We conducted exhaustive searches across premier academic databases and preprint repositories to ensure broad coverage of both peer-reviewed and cutting-edge unpublished works. The primary databases included Google Scholar, DBLP, Web of Science, Scopus, and arXiv.

To construct a robust search query, we combined terms related to the financial domain, time-series data, and generative tasks. The primary Boolean search strings included: (*"financial time-series generation"* OR *"synthetic financial data"* OR *"market simulation generative model"*) AND (*"financial diffusion model"* OR *"financial GAN"* OR *"time-series imputation finance"* OR *"financial foundation models"*).

Given the rapid evolution of deep generative models, we restricted our primary search window to papers published or uploaded between 2021 and early 2026 (including early-access and preprints). We included articles from high-impact journals (e.g., *Information Sciences*, *Expert Systems with Applications*, *Quantitative Finance*), top-tier computer science conferences (e.g., NeurIPS, ICML, ICLR, KDD, IJCAI), and highly cited arXiv preprints that represent significant methodological breakthroughs.

After full-text screening, the final corpus comprises 131 studies, each assigned to a unique primary task category: extrapolation (54), imputation (29), and synthesis (47). Multi-task or cross-cutting contributions are discussed in the relevant sections but are not double-counted in task-level comparison tables.

Inclusion and exclusion criteria

As illustrated in Fig. 6, all retrieved papers underwent a two-stage screening process, comprising title/abstract screening followed by full-text review against explicit eligibility criteria.

Studies were eligible when they explicitly formulated or addressed the generation, imputation, or fully synthetic simulation of financial time-series data, proposed novel generative architectures specifically adapted for financial contexts, or established unified evaluation protocols, benchmarks, or datasets for synthetic financial data.

We excluded purely predictive or forecasting studies that treated outputs strictly as deterministic point estimates without a generative, distributional, or probabilistic perspective. We also excluded general time-series generation papers without an identifiable and empirically tested financial application, such as studies tested solely on weather or traffic data, together with earlier surveys or methodologies that did not incorporate modern deep learning or generative AI techniques unless they were necessary for historical context.

Data extraction and coding protocol

Two reviewers independently screened records and completed a structured extraction form; disagreements were resolved by discussion, consistent with PRISMA 2020 guidance[29].

Task type recorded whether a study addressed extrapolation, imputation, micro-level synthesis, or macro-level market simulation. Model family classified each method as statistical or probabilistic, discriminative deep learning, GAN-based, diffusion or score-based, or foundation-model or LLM-based. Data modality captured whether the study used univariate or multivariate series, its sampling frequency, and any conditioning variables such as text or macroeconomic indicators. Evaluation metrics documented the quantitative and qualitative measures used to validate generated outputs, including MSE, Wasserstein distance, stylized facts, and downstream utility. Code and data availability recorded whether authors released code or used public benchmark datasets.

Review dimensions and taxonomy synthesis

The extracted data were synthesized in the Results and Discussion sections along six analytical dimensions.

Task formulation defined the mathematical boundaries and generative objectives of extrapolation, imputation, and synthesis, as presented in Problem Formulation and Setup. Model design examined how specific generative architectures address financial data challenges across the extrapolation, imputation, and synthesis subsections. Applications examined how these tasks support downstream quantitative workflows. Evaluation protocol assessed how the financial realism and empirical validity of generated series are quantified. Datasets and benchmarks summarized the empirical foundations used to train and evaluate generative models. Open challenges and the roadmap identified theoretical and practical bottlenecks requiring future research and are discussed in the Discussion section. By rigorously adhering to this methodology, this survey transitions from a mere enumeration of literature to a systematic, evaluation-oriented resource paper, providing a reproducible basis for subsequent synthesis and critical appraisal.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to the fact that the literature retrieval query logs and the systematic extraction coding tables contain internal reviewer annotations and are subject to licensing restrictions on the underlying subscription databases, but they are available from the corresponding author on reasonable request. All benchmark datasets, curated databases, and public APIs reviewed and analyzed in this survey (including FinTSB, TFB, TSGBench, CTBench, FinTSBridge, WRDS, CRSP, and FRED) are derived from published literature and publicly accessible repositories as referenced in the Data Resources and Benchmarks subsection.

Code availability

This work is a systematic literature survey and did not generate custom algorithmic codebases or software artifacts. All generative models, benchmark evaluation suites, and foundational model repositories discussed are publicly accessible via their respective cited publications and open-source repositories as cataloged throughout the manuscript.

Received: 17 June 2026; Accepted: 21 September 2026;
Published online: 19 October 2026

References

1. Hamilton, J. D. Time Series Analysis (Princeton University Press, Princeton, NJ, 1994).
2. Box, G. E., Jenkins, G. M., Reinsel, G. C. & Ljung, G. M. Time series analysis: forecasting and control (John Wiley & Sons, 2015).
3. Campbell, J. Y., Lo, A. W. & MacKinlay, A. C. The Econometrics of Financial Markets (Princeton University Press, Princeton, NJ, 1997).
4. Tsay, R. S. Analysis of Financial Time Series (Wiley, Hoboken, NJ, 2010).
5. Hyndman, R. J. & Athanasopoulos, G. Forecasting: Principles and Practice (OTexts, Melbourne, Australia, 3 edn, 2021).
6. Gneiting, T. & Katzfuss, M. Probabilistic forecasting. *Annu. Rev. Stat. Its Application* **1**, 125–151, https://doi.org/10.1146/annurev-statistics-062713-085831 (2014).
7. Yoon, J., Jarrett, D. & van der Schaar, M. Time-series generative adversarial networks. In *NeurIPS* (2019).
8. Wiese, M., Knobloch, R., Korn, R. & Muller, P. QuantGANs: deep generation of financial time series. *Quant. Financ.* **20**, 1419–1440 (2020).
9. Naritomi, Y. & Adachi, T. Data augmentation of high frequency financial data using generative adversarial network. In *Proc. of the IEEE/WIC/ACM International Conference on Web Intelligence (WI/IAT)* (2020).
10. Yoon, J., Jordon, J. & van der Schaar, M. GAIN: Missing data imputation using generative adversarial nets. In *Proc. of the International Conference on Machine Learning (ICML)* (2018).
11. Tashiro, Y., Song, J., Song, Y. & Ermon, S. CSDI: Conditional score-based diffusion models for probabilistic time series imputation. In *Adv. in Neural Inform. Proces. Sys.* **34**, 24804–24816 (2021).
12. Wang, S. et al. CWGAIN-GP: Conditional wasserstein GAN with gradient penalty for contiguous missing data. *Quant. Finance* (2024).
13. Rasul, K., Seward, C., Nouretdinov, I. & Vollgraf, R. Autoregressive denoising diffusion models for multivariate probabilistic time series forecasting. In *International Conference on Machine Learning (ICML)*, 8857–8868 (PMLR, 2021).
14. Cont, R. Empirical properties of asset returns: stylized facts and statistical issues. *Quant. Financ.* **1**, 223–236 (2001).
15. Goodfellow, I. et al. Generative adversarial nets. In *Adv. Neural Inform. Process. Sys.* **27** (2014).
16. Ho, J., Jain, A. & Abbeel, P. Denoising diffusion probabilistic models. In *Adv. Neural Inform. Proces. Sys. (NeurIPS)*, vol. 33, 6840–6851 (2020).
17. Vaswani, A. et al. Attention is all you need. In *Adv. Neural Inform. Process. Sys.* **30** (2017).
18. Brown, T. B. et al. Language models are few-shot learners. In *Adv. Neural Inform. Proces. Syst.***33**, 1877–1901 (2020).
19. Jin, M. et al. Time-LLM: Time series forecasting by reprogramming large language models. In *Int. Conference on Learning Representations (ICLR)* (2024).
20. Ansari, A. F. et al. Chronos: Learning the language of time series. *arXiv preprint*. 2403.07815 (2024).
21. Zhang, X., Chowdhury, R. R., Gupta, R. K. & Shang, J. Large language models for time series: A survey. In *IJCAI* (2024).
22. Liang, Y. et al. Foundation models for time series analysis: A tutorial and survey. In *KDD*, https://doi.org/10.1145/3637528.3671451 (2024).
23. Kim, J., Kim, H., Kim, H., Lee, D. & Yoon, S. A comprehensive survey of deep learning for time series forecasting: architectural diversity and open challenges. *Artif. Intell. Rev*.https://doi.org/10.1007/S10462-025-11223-9 (2025).
24. Kong, X. et al. Deep learning for time series forecasting: a survey. *Int. J. Mach. Learn. Cybern*.https://doi.org/10.1007/S13042-025-02560-W (2025).

25. Nie, Y. et al. A survey of large language models for financial applications: Progress, prospects and challenges. *CoRR*. 2406.11903 (2024).
26. Arsenault, P., Wang, S. & Patenaude, J. A survey of explainable artificial intelligence (XAI) in financial time series forecasting. *ACM Comput. Surv*.https://doi.org/10.1145/3729531 (2025).
27. Hu, Y., Liu, P., Zhu, Y. & Jiang, C. FinTSB: A comprehensive and practical benchmark for financial time series forecasting. 2502.18834 (2025).
28. Qiu, X. et al. TFB: towards comprehensive and fair benchmarking of time series forecasting methods. *Proc. VLDB Endow*.https://doi.org/10.14778/3665844.3665863 (2024).
29. Page, M. J. et al. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ* **372**, n71 https://doi.org/10.1136/bmj.n71 (2021).
30. Box, G. E. P. & Jenkins, G. M. Time series analysis: Forecasting and control (Holden-Day, 1970).
31. Bollerslev, T. Generalized autoregressive conditional heteroskedasticity. *J. Econ.* **31**, 307–327, https://doi.org/10.1016/0304-4076(86)90063-1 (1986).
32. Elman, J. L. Finding structure in time. *Cogn. Sci.* **14**, 179–211, https://doi.org/10.1207/s15516709cog1402_1 (1990).
33. Hochreiter, S. & Schmidhuber, J. Long short-term memory. *Neural Comput.* **9**, 1735–1780, https://doi.org/10.1162/neco.1997.9.8.1735 (1997).
34. Arjovsky, M., Chintala, S. & Bottou, L. Wasserstein generative adversarial networks. In *Proc. of the 34th International Conference on Machine Learning*, vol. 70 of *Proceedings of Machine Learning Research*, 214–223 (2017).
35. Gulrajani, I., Ahmed, F., Arjovsky, M., Dumoulin, V. & Courville, A. C. Improved training of Wasserstein GANs. In *Advances in Neural Information Processing Systems*, vol. 30 (2017).
36. Song, Y. & Ermon, S. Generative modeling by estimating gradients of the data distribution. In *Advances in Neural Information Processing Systems*, vol. 32 (2019).
37. Song, Y. et al. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations* (2021).
38. Zhou, T., Niu, P., Wang, X., Sun, L. & Jin, R. One fits all: Power general time series analysis by pretrained LM. In *Adv. Neural Information Process. Sys. (NeurIPS)* (2023).
39. Narasimhan, S. & Jiang, W. Time weaver: Specificity in conditional time series generation. *arXiv preprint*. 2403.02682 (2024).
40. Ge, Y., Li, J., Zhao, Y., Wen, H. & Pan, S. T2S: high-resolution time series generation with text-to-series diffusion models. In *IJCAI*, https://doi.org/10.24963/IJCAI.2025/580 (2025).
41. Lozano, C., Bagén, E. & Vives, R. Dual-sourced diffusion models for coherent financial market synthesis. *Quant. Financ.* **23**, 1605–1620 (2023).
42. Li, Q., Kamaruddin, N., Yuhaniz, S. S. & Al-Jaifi, H. A. A. Forecasting stock prices changes using long-short term memory neural network with symbolic genetic programming. *Sci. Rep.* **14**, 422 https://doi.org/10.1038/s41598-023-50783-0 (2024).
43. Zhu, Z., Chen, H., Qu, Q. & Chung, V. FinCast: A foundation model for financial time-series forecasting. *arXiv preprint*. 2508.19609 (2025).
44. Gneiting, T. & Raftery, A. E. Strictly proper scoring rules, prediction, and estimation. *J. Am. Stat. Assoc.* **102**, 359–378 https://doi.org/10.2198/10.1198/016214506000001437 (2007).
45. Zhou, Y., Shang, X., Sylla, I. & Yu, H. RegTensor: Fast non-linear coupled tensor completion with orthogonal regularization. *Knowl.-Based Syst*. (2023).
46. Cao, D., Ye, W., Zhang, Y. & Liu, Y. TimeDiT: General-purpose diffusion transformers for time series foundation model. *arXiv preprint*. 2409.02322 (2024).
47. Tanaka, Y., Takayanagi, T., Piao, Z. & Izumi, K. CoFinDiff: Controllable financial diffusion model for time series generation. In *Proc. of the International Joint Conference on Artificial Intelligence (IJCAI)* (2025).
48. Gneiting, T., Stanberry, L. I., Jentsch, T. & Dean, A. G. Assessing probabilistic forecasts of multivariate quantities, with an application to ensemble predictions of surface winds. *J. R. Stat. Soc.: Ser. B* **70**, 635–652 https://doi.org/10.1111/j.1467-9868.2007.00652.x (2008).
49. Nelson, D. B. Conditional heteroskedasticity in asset returns: A new approach. *Econometrica* **59**, 347–370 https://doi.org/10.2307/2938260 (1991).
50. Qin, A., Li, B., Cai, C. & Ding, D. Advances in LSTM architectures for financial sequence modeling. *Expert Syst. Appl*. (2025).
51. Li, J.-C., Sun, L.-P., Wu, X. & Tao, C. Enhancing financial time series forecasting with hybrid deep learning: CEEMDAN-Informer-LSTM model. *Appl. Soft Comput.* **177**, 113241 (2025).
52. Çalışkan, B. Enhancing recurrent neural networks for stock market forecasts through PEC-W framework. In *2025 IEEE Symposium on Computational Intelligence for Financial Engineering and Economics (CIFEr)* (IEEE, 2025).
53. Mallat, S. G. A theory for multiresolution signal decomposition: The wavelet representation. *IEEE Trans. Pattern Anal. Mach. Intell.* **11**, 674–693 https://doi.org/10.1109/34.192463 (1989).
54. Torres, M. E., Colominas, M. A., Schlotthauer, G. & Flandrin, P. A complete ensemble empirical mode decomposition with adaptive noise. In *2011 IEEE International Conference on Acoustics, Speech and Signal Processing*, 4144–4147 https://doi.org/10.1109/ICASSP.2011.5947265 (2011).
55. Zhou, H. et al. Informer: Beyond efficient transformer for long sequence time-series forecasting. In *Proc. of the AAAI Conference on Artificial Intelligence*, vol. 35, 11106–11115 https://doi.org/10.1609/aaai.v35i12.17325 (2021).
56. Koza, J. R. Genetic Programming: On the Programming of Computers by Means of Natural Selection (MIT Press, 1992).
57. Arjovsky, M., Bottou, L., Gulrajani, I. & Lopez-Paz, D. Invariant risk minimization. *arXiv preprint*. 1907.02893 (2019).
58. Cao, D. et al. Large scale financial time series forecasting with multi-faceted model. In *Proc. of the Fourth ACM International Conference on AI in Finance*, 472–480 https://doi.org/10.1145/3604237.3626868 (2023).
59. Dai, Z., Xinyue, Z. & Youchen, W. WaveGAN: Wavelet-based generative adversarial networks for financial time series. *Quant. Finance* (2024).
60. Dai, A., Filho, B., Sclip, C. & Chattoraj, D. N-BEATS-GAN: Interpretable generation of financial sequences. *Expert Syst. Appl*. (2026).
61. Oreshkin, B. N., Carpov, D., Chapados, N. & Bengio, Y. N-BEATS: Neural basis expansion analysis for interpretable time series forecasting. In *International Conference on Learning Representations (ICLR)* (2020).
62. Peebles, W. & Xie, S. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 4195–4205 https://doi.org/10.1109/ICCV51070.2023.00389 (2023).
63. Wang, S., Wang, T., Li, S. et al. StockTime: Multi-modal fusion for financial forecasting. *arXiv preprint* (2024). 2409.08281.
64. Liu, Z. & Jia, R. LLM4FTS: Large language models for financial time series. *arXiv preprint* (2025). 2505.02880.
65. Tang, J., Gong, X., Zhang, Z. & Tao, M. LLM-PS: Time-to-text translation for financial modeling. *arXiv preprint* (2025). 2503.09656.
66. Sakoe, H. & Chiba, S. Dynamic programming algorithm optimization for spoken word recognition. *IEEE Trans. Acoust., Speech, Signal Process.* **26**, 43–49 https://doi.org/10.1109/TASSP.1978.1163055 (1978).

67. Liu, K., Dai, L., Li, M. & Xia, N. CALF: Consistency alignment for language and finance. *arXiv preprint* (2025).
68. Qi, Y., Lei, P., Zhang, Q. & Shen, R. TimeHF: Human feedback for time series foundation models. *arXiv preprint* (2025). 2501.15942.
69. Christiano, P. F. et al. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems*, vol. 30 (2017).
70. Ouyang, L. et al. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, vol. 35, 27730–27744 (2022).
71. Shazeer, N. et al. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *International Conference on Learning Representations* (2017).
72. Fedus, W., Zoph, B. & Shazeer, N. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *J. Mach. Learn. Res.* **23**, 1–39 (2022).
73. Koenker, R. & Bassett, G. Regression quantiles. *Econometrica* **46**, 33–50, https://doi.org/10.2307/1913643 (1978).
74. Hinton, G., Vinyals, O. & Dean, J. Distilling the knowledge in a neural network. In *NeurIPS Deep Learning and Representation Learning Workshop*. ArXiv:1503.02531 (2015).
75. Sharpe, W. F. The sharpe ratio. *J. Portf. Manag.* **21**, 49–58, https://doi.org/10.3905/jpm.1994.409501 (1994).
76. O'Hara, M. Market Microstructure Theory (Blackwell, 1995).
77. Gould, M. D. et al. Limit order books. *Quant. Financ.* **13**, 1709–1742, https://doi.org/10.1080/14697688.2013.803148 (2013).
78. McCloskey, M. & Cohen, N. J. Catastrophic interference in connectionist networks: The sequential learning problem. *Psychol. Learn. Motiv.* **24**, 109–165 (1989).
79. Zenke, F., Poole, B. & Ganguli, S. Continual learning through synaptic intelligence. In *International Conference on Machine Learning (ICML)*, 3987–3995 (PMLR, 2017).
80. Rubin, D. B. Inference and missing data. *Biometrika* **63**, 581–592, https://doi.org/10.1093/biomet/63.3.581 (1976).
81. Hayashi, T. & Yoshida, N. On a covariance estimator of separately observed diffusions. *Appl. Math. Financ.* **12**, 355–381, https://doi.org/10.1080/13504860500141201 (2005).
82. Candès, E. J. & Recht, B. Exact matrix completion via convex optimization. *Found. Comput. Math.* **9**, 717–772, https://doi.org/10.1007/s10208-009-9045-5 (2009).
83. Chow, G. C. & Lin, A.-l. Best linear unbiased interpolation, distribution, and extrapolation of time series by related series. *Rev. Econ. Stat*. 372–375 (1971).
84. Tibshirani, R. Regression shrinkage and selection via the Lasso. *J. R. Stat. Soc.: Ser. B* **58**, 267–288, https://doi.org/10.1111/j.2517-6161.1996.tb02080.x (1996).
85. Troyanskaya, O. et al. Missing value estimation methods for DNA microarrays. *Bioinformatics* **17**, 520–525, https://doi.org/10.1093/bioinformatics/17.6.520 (2001).
86. van Buuren, S., Boshuizen, H. C. & Knook, D. L. Multiple imputation of missing blood pressure covariates in survival analysis. *Stat. Med.* **18**, 681–694 (1999).
87. Breiman, L. Random forests. *Mach. Learn.* **45**, 5–32, https://doi.org/10.1023/A:1010933404324 (2001).
88. Stekhoven, D. J. & Bühlmann, P. MissForest-non-parametric missing value imputation for mixed-type data. *Bioinformatics* **28**, 112–118 (2012).
89. Khuwaja, P., Khowaja, S. A. & Dev, K. Adversarial learning networks for FinTech applications using heterogeneous data sources. *IEEE Internet Things J.*https://doi.org/10.1109/JIOT.2021.3100742 (2023).
90. Saunders, C., Gammerman, A. & Vovk, V. Ridge regression learning algorithm in dual variables. In *Proceedings of the 15th International Conference on Machine Learning*, 515–521 (1998).
91. Aronszajn, N. Theory of reproducing kernels. *Trans. Am. Math. Soc.* **68**, 337–404, https://doi.org/10.2307/1990404 (1950).
92. Smith, A. Discount curve estimation with kernel ridge regression. *J. Financ. Econom*. (2020).
93. Sideras, A., Bougiatiotis, K., Zavitsanos, E., Paliouras, G. & Vouros, G. Bankruptcy prediction: Data augmentation, LLMs and the need for auditor's opinion. In *Proceedings of the 5th ACM International Conference on AI in Finance* (2024).
94. Wang, X. et al. Multivariate time series imputation using BiGRU-BiLSTM and polynomial regression. *Expert Syst. Appl*. (2021).
95. Cho, K. et al. Learning phrase representations using RNN encoder–decoder for statistical machine translation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing*, 1724–1734 https://doi.org/10.3115/v1/D14-1179 (2014).
96. Schuster, M. & Paliwal, K. K. Bidirectional recurrent neural networks. *IEEE Trans. Signal Process.* **45**, 2673–2681 https://doi.org/10.1109/78.650093 (1997).
97. Ivakhnenko, A. G. Polynomial theory of complex systems. *IEEE Trans. Syst., Man, Cybern.* **SMC-1**, 364–378 https://doi.org/10.1109/TSMC.1971.4308320 (1971).
98. Varolgunes, M., Zhou, Y., Yu, H. & Uddin, M. NMTucker: Non-linear matryoshka tucker decomposition for tensor completion. *Inf. Sci*. (2023).
99. Tucker, L. R. Some mathematical notes on three-mode factor analysis. *Psychometrika* **31**, 279–311 https://doi.org/10.1007/BF02289464 (1966).
100. Wang, L., Cui, Y., Guo, Z. & Ning, P. AttnWGAIN: Attention-based wasserstein GAN for missing financial data imputation. *Expert Syst. Appl*. (2025).
101. Qin, J. & Wang, Q. ImputeGAN: Informer-based generative adversarial network for long-sequence time-series imputation. *Appl. Soft Comput*. (2023).
102. Liang, G. et al. Latent space score-based diffusion model for probabilistic multivariate time series imputation. In *ICASSP*, https://doi.org/10.1109/ICASSP49660.2025.10888912 (2025).
103. Zhang, S. et al. Score-CDM: Score-weighted convolutional diffusion model for multivariate time series imputation. In *IJCAI* (2024).
104. Zhang, S. et al. SaSDim: Self-adaptive noise scaling diffusion model for spatial time series imputation. In *IJCAI* (2024).
105. Zhang, Y., Tan, X., Wang, L. & Wang, J. Spatial-temporal consistency in diffusion models for imputation. *IEEE Trans. Knowl. Data Eng*. (2024).
106. Zhou, H., Zheng, Y., Wang, Q. & Zhou, T. Cross-variable dependency alignment in financial diffusion imputation. *Inf. Sci*. (2024).
107. Fons, E. et al. LSCD: Lomb-scargle conditioned diffusion for time series imputation. *arXiv preprint* (2025). 2506.17039.
108. Zhang, Q., Wang, R., Qiu, H., Liu, X. & Cheng, X. SPDM: spatiotemporal-periodic diffusion model for multivariate time series imputation. In *CSCWD*, https://doi.org/10.1109/CSCWD64889.2025.11033512 (2025).
109. He, X. et al. LLM-Forest: Ensemble learning of LLMs with graph-augmented prompts for data imputation. *arXiv preprint* (2024). 2410.21520.
110. Liu, Z. et al. MTabGen: Mixed-type tabular data generation and imputation via diffusion models. *arXiv preprint* (2023).
111. Shan, N., Li, J. & Oliva, P. Deep learning for irregular and missing financial data: A review. *IEEE Access* (2021).
112. Du, Y. et al. A comprehensive survey on deep learning for time-series imputation. *IEEE Trans. Pattern Anal. Mach. Intell*. (2024).
113. Miao, P., Li, C. & Chen, X. Joint forecasting and imputation via coupled diffusion models. *Appl. Soft Comput*. (2025).
114. Zhang, Y. et al. Robust financial imputation via dual-attention networks. *Inf. Sci*. (2023).

115. Cai, Y. et al. Temporal graph networks for financial time-series imputation. *Knowl.-Based Syst*. (2025).
116. Rossi, E. et al. Temporal graph networks for deep learning on dynamic graphs. In *ICLR 2020 Workshop on Representation Learning on Graphs and Manifolds* (2020).
117. Chen, L. et al. Imputation strategies for fragmented financial markets. *Knowl.-Based Syst*. (2023).
118. Zhang, J. et al. Advancing multi-modal imputation in quantitative finance. *IEEE Trans. Knowl. Data Eng*. (2025).
119. Bamford, S., Fons, E., El-Laham, N. & Vyetrenko, S. High-frequency data imputation using deep learning and order book dynamics. *Quant. Finance* (2023).
120. Shankar, A., Chen, Z., van Deursen, A. & Hai, R. Benchmarking imputation algorithms for limit order book data. *Quant. Finance* (2025).
121. Nater, C. & Khayati, M. Scalable probabilistic imputation for high-frequency trading systems. *Inf. Syst*. (2025).
122. Qian, X., Zhang, L., Zhao, P. & Cheng, K. Dynamic regime-aware imputation for multivariate time series. *Inf. Sci*. (2024).
123. Gao, R. et al. Score-based models for non-stationary market data imputation. *Expert Syst. Appl*. (2024).
124. Wi, J., Shin, H. & Park, J. Generative imputation for financial stress testing. *Expert Syst. Appl*. (2023).
125. Wang, H., Yang, Y., Sun, C., Wen, X. & Wang, K. Evaluating look-ahead bias in deep learning based financial imputation. *J. Financ. Econom*. (2024).
126. Mandelbrot, B. The variation of certain speculative prices. *J. Bus.* **36**, 394–419, https://doi.org/10.1086/294632 (1963).
127. Bai, S., Kolter, J. Z. & Koltun, V. An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. *arXiv preprint arXiv:1803.01271* (2018).
128. Huang, A., Khushi, M. & Suleiman, B. Regime-specific quant generative adversarial network: A conditional generative adversarial network for regime-specific deepfakes of financial time series. *Appl. Sci*. (2023).
129. Rabiner, L. R. A tutorial on hidden Markov models and selected applications in speech recognition. *Proc. IEEE* **77**, 257–286 https://doi.org/10.1109/5.18626 (1989).
130. Hamilton, J. D. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica* **57**, 357–384 https://doi.org/10.2307/1912559 (1989).
131. Sun, H., Deng, Z., Chen, H. & Parkes, D. C. Decision-aware conditional GANs for time series data. In *ICAIF*, https://doi.org/10.1145/3604237.3626855 (2023).
132. Rizzato, M., Wallart, J., Geissler, C., Morizet, N. & Boumlaik, N. Generative adversarial networks applied to synthetic financial scenarios generation. *Physica A: Statistical Mechanics and its Applications* (2023).
133. Cont, R., Mihai, C., Renyuan, X. & Chao, Z. Tail-GAN: Learning to simulate tail risk scenarios. *Manage. Sci*. (2025).
134. Acerbi, C. & Tasche, D. Expected shortfall: A natural coherent alternative to value at risk. *Economic Notes* **31**, 379–388, https://doi.org/10.1111/1468-0300.00091 (2002).
135. Acerbi, C. & Székely, B. Back-testing expected shortfall. *Risk* **27**, 76–81 (2014).
136. Takahashi, T. & Mizuno, T. Generation of synthetic financial time series by diffusion models (2024). 2410.18897.
137. Kim, G., Choi, S. & Kim, Y. A diffusion-based generative model for financial time series via geometric brownian motion (2025). 2507.19003.
138. Wang, Z. & Ventre, C. A financial time series denoiser based on diffusion models. In *Proc. of the ACM International Conference on AI in Finance (ICAIF)* (2024).
139. Black, F. & Scholes, M. The pricing of options and corporate liabilities. *J. Political Econ.* **81**, 637–654, https://doi.org/10.1086/260062 (1973).
140. Masi, G., Conti, M., Bartolini, N. & Vyetrenko, S. On correlated stock market time series generation. In *ICAIF*, https://doi.org/10.1145/3604237.3626895 (2023).
141. Istiaque, R. A., Pun, C. S. & Song, Y. Simulating asset prices using conditional time-series GAN. In *Proc. of the ACM International Conference on AI in Finance (ICAIF)* (2024).
142. Xia, H., Wang, J., Zhang, L. & Chen, W. Market-GAN: A two-stage GAN-based framework for controllable financial market generation (2024).
143. Kazadaev, M., Pozdnyakov, V. & Makarov, I. Time series generation with GANs for momentum effect simulation on moscow stock exchange. In *CIFEr*, https://doi.org/10.1109/CIFER62890.2024.10772763 (2024).
144. Rusnak, A. M. & Daul, S. Macroeconomic conditioned synthetic financial markets. In *Proceedings of the ACM International Conference on AI in Finance (ICAIF)* (2024).
145. Berti, L., Prenkaj, B. & Velardi, P. TRADES: Generating realistic market simulations with diffusion models (2025). 2502.07071.
146. Huang, Y.-H., Liu, Y., Liu, W. & Bian, J. Controllable financial market generation with diffusion guided meta agent (2024). 2408.12991.
147. Li, J., Liu, W., Fang, S. & Bian, J. MarS: A financial market simulation engine powered by generative foundation model. In *Proc. of the International Conference on Learning Representations (ICLR)* (2025).
148. Zhang, F., Wang, L. & Zhang, X. Desensitized financial data generation based on generative adversarial network and differential privacy. *Big Data Min. Anal*.https://doi.org/10.26599/BDMA.2024.9020047 (2025).
149. Dwork, C. Differential privacy. In *Automata, Languages and Programming*, vol. 4052 of *Lecture Notes in Computer Science*, 1–12, https://doi.org/10.1007/11787006_1 (Springer, 2006).
150. Dwork, C. & Roth, A. The Algorithmic Foundations of Differential Privacy (Now Foundations and Trends, 2014).
151. Shokri, R., Stronati, M., Song, C. & Shmatikov, V. Membership inference attacks against machine learning models. In *2017 IEEE Symposium on Security and Privacy*, 3–18 https://doi.org/10.1109/SP.2017.41 (2017).
152. Gholamrezaei, M., Koochali, A., Dengel, A. & Ahmed, S. Conditioning financial generation on alternative metadata streams. *Knowl. -Based Syst.* **270**, 110531 (2023).
153. Tao, R. et al. Efficient generation of financial time series via diffusion distillation. *arXiv preprint* (2024). 2406.11223.
154. Oppel, J. et al. Practical stress testing with conditional generative models. *J. Risk* (2025).
155. Liu, X.-Y. et al. FinRL: A deep reinforcement learning library for automated stock trading in quantitative finance. In *NeurIPS Workshop on Deep Reinforcement Learning* (2020).
156. Liu, X.-Y. et al. FinRL-Meta: Market environments and benchmarks for data-driven financial reinforcement learning. In *Adv. Neural Inform. Processing Sys. (NeurIPS)* (2022).
157. Schulman, J., Wolski, F., Dhariwal, P., Radford, A. & Klimov, O. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347* (2017).
158. Mnih, V. et al. Asynchronous methods for deep reinforcement learning. In *Proc. of the 33rd International Conference on Machine Learning*, vol. 48 of *Proc. of Machine Learning Research*, 1928–1937 (2016).
159. Haarnoja, T., Zhou, A., Abbeel, P. & Levine, S. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *Proc. of the 35th International Conference on Machine Learning*, vol. 80 of *Proceedings of Machine Learning Research*, 1861–1870 (2018).

160. Wang, R., Wei, H., An, B., Feng, Z. & Yao, J. Deep stock trading: A hierarchical reinforcement learning framework for portfolio optimization and order execution. In *Proc. of the AAAI Conference on Artificial Intelligence* (2020).
161. Boyle, P. P. Options: A monte carlo approach. *J. Financial Econ.* **4**, 323–338, https://doi.org/10.1016/0304-405X(77)90005-8 (1977).
162. Lu, J. & Ding, D. A hybrid approach on conditional GAN for portfolio analysis (2022). 2208.07159.
163. Markowitz, H. Portfolio selection. *J. Financ.* **7**, 77–91, https://doi.org/10.2307/2975974 (1952).
164. Herculano, M. C. & Punnoose, J. Financial condition indices in an incomplete data environment. *Stud. Nonlinear Dyn. Econom*. (2025).
165. Emaan, K. U. Improving credit card fraud detection through transformer-enhanced GAN oversampling (2025). 2509.19032.
166. Chawla, N. V., Bowyer, K. W., Hall, L. O. & Kegelmeyer, W. P. SMOTE: Synthetic minority over-sampling technique. *J. Artif. Intell. Res.* **16**, 321–357 https://doi.org/10.2514/1.9530 (2002).
167. Liu, X., Zhang, Z. & Wang, D. Classification of imbalanced credit scoring data sets based on ensemble method with the weighted-hybrid-sampling (2021).
168. Byrd, D., Hybinette, M. & Balch, T. H. ABIDES: Towards high-fidelity market simulation for AI research (2019). 1904.12066.
169. Amrouni, S. et al. ABIDES-Gym: Gym environments for multi-agent discrete event simulation and application to financial markets. In *Proc. of the ACM International Conference on AI in Finance (ICAIF)* (2021).
170. Brockman, G. et al. OpenAI Gym. *arXiv preprint arXiv:1606.01540* (2016).
171. Balch, T. H., Mahfouz, M., Lockhart, J., Hybinette, M. & Byrd, D. How to evaluate trading strategies: Single agent market replay or multiple agent interactive simulation? (2019). 1906.12010.
172. Rollins, M. & Cliff, D. Which trading agent is best? using a threaded parallel simulation of a financial market changes the pecking-order (2020).
173. Politis, D. N. & Romano, J. P. The stationary bootstrap. *J. Am. Stat. Assoc.* **89**, 1303–1313, https://doi.org/10.1080/01621459.1994.10476870 (1994).
174. Maillard, S., Roncalli, T. & Teïletche, J. The properties of equally weighted risk contribution portfolios. *J. Portf. Manag.* **36**, 60–70 https://doi.org/10.3905/jpm.2010.36.4.060 (2010).
175. Glosten, L. R. & Milgrom, P. R. Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *J. Financial Econ.* **14**, 71–100 https://doi.org/10.1016/0304-405X(85)90044-3 (1985).
176. Huang, R. D. & Stoll, H. R. Dealer versus auction markets: A paired comparison of execution costs on NASDAQ and the NYSE. *J. Financial Econ.* **41**, 313–357 https://doi.org/10.1016/0304-405X(95)00873-F (1996).
177. Gao, Y., Lui, K. Y. C. & Hernandez-Leal, P. Robust risk-sensitive reinforcement learning agents for trading markets. In *Proc. of the ACM International Conference on AI in Finance (ICAIF)* (2021).
178. Balch, T., Potluru, V. K., Paramanand, D. & Veloso, M. Six levels of privacy: A framework for financial synthetic data. In *Proc. the ACM International Conference on AI in Finance (ICAIF)* (2024).
179. Boudewijn, A. et al. Privacy measurement in tabular synthetic data: State of the art and future research directions. In *Proc. Privacy Enhancing Technologies (PETS)* (2023).
180. Ang, Y., Bao, Y., Tung, A. K. H. & Huang, Z. TSGBench: Time series generation benchmark. *Proc. VLDB Endow.* **17**, 305–318 (2024).
181. Ang, Y., Huang, Q., Bao, Y. & Huang, Z. CTBench: Cryptocurrency time series generation benchmark (2025). 2508.02758.
182. Wang, Y., Gao, T., Zhang, H. & Zhang, X. FinTSBridge: A new evaluation suite for real-world financial prediction with advanced time series models (2025). 2503.06928.
183. Heusel, M. et al. GANs trained by a two time-scale update rule converge to a local Nash equilibrium. In *Adv. Neural Information Proc. Sys.***30** (2017).
184. Villani, C. Optimal Transport: Old and New (Springer, 2009).
185. Gretton, A., Borgwardt, K. M., Rasch, M. J., Schölkopf, B. & Smola, A. A kernel two-sample test. *J. Mach. Learn. Res.* **13**, 723–773 (2012).
186. Cooley, J. W. & Tukey, J. W. An algorithm for the machine calculation of complex fourier series. *Math. Comput.* **19**, 297–301, https://doi.org/10.2307/2003354 (1965).
187. Hill, B. M. A simple general approach to inference about the tail of a distribution. *Ann. Stat.* **3**, 1163–1174, https://doi.org/10.1214/aos/1176343247 (1975).
188. Black, F. Studies of stock price volatility changes. In *Proc. of the 1976 Meetings of the American Statistical Association, Business and Economics Statistics Section*, 177–181 (American Statistical Association, 1976).
189. Esteban, C., Hyland, S. L. & Rätsch, G. Real-valued (medical) time series generation with recurrent conditional GANs. *arXiv preprint arXiv:1706.02633* (2017).
190. Bindschaedler, V., Shokri, R. & Gunter, C. A. Plausible deniability for privacy-preserving data synthesis. *Proc. VLDB Endow.* **10**, 541–552 https://doi.org/10.14778/3055540.3055547 (2017).
191. Ljung, G. M. & Box, G. E. P. On a measure of lack of fit in time series models. *Biometrika* **65**, 297–303 https://doi.org/10.1093/biomet/65.2.297 (1978).
192. Zhong, M., Lin, Y., Yang, P. & Chen, L. Representation learning of limit order book: A comprehensive study and benchmarking. *arXiv preprint* (2024). 2412.01802.
193. Jiang, J. et al. FinMaster: A holistic benchmark for mastering full-pipeline financial workflows with llms. *arXiv preprint* (2024). 2412.18025.
194. Yu, F. et al. Tsrbench: A comprehensive multi-task multi-modal time series reasoning benchmark for generalist models. *arXiv preprint* (2025). 2502.13840.
195. Tetlock, P. C. Giving content to investor sentiment: The role of media in the stock market. *J. Financ.* **62**, 1139–1168 https://doi.org/10.1111/j.1540-6261.2007.01232.x (2007).
196. Loughran, T. & McDonald, B. When is a liability not a liability? textual analysis, dictionaries, and 10-ks. *J. Financ.* **66**, 35–65 https://doi.org/10.1111/j.1540-6261.2010.01625.x (2011).
197. Wharton Research Data Services. WRDS: Wharton research data services. Online research database, The Wharton School, University of Pennsylvania. Accessed: 2025-09-10 (2025).
198. Center for Research in Security Prices. CRSP: Center for research in security prices. Online research database, Booth School of Business, University of Chicago. Accessed: 2025-09-10 (2025).
199. S&P Global Market Intelligence. Compustat: Fundamentals and market data. Online financial database, S&P Global. Accessed: 2025-09-10 (2025).
200. Brown, S. J., Goetzmann, W., Ibbotson, R. G. & Ross, S. A. Survivorship bias in performance studies. *Rev. Financial Stud.* **5**, 553–580 https://doi.org/10.59629/rfs/5.4.553 (1992).
201. Federal Reserve Bank of St. Louis. FRED: Federal reserve economic data. Online macroeconomic database. Accessed: 2025-09-10 (2025).
202. CSMAR. CSMAR: China stock market and accounting research database. Online research database, GTA Information Technology Co. Ltd. Accessed: 2025-09-10 (2025).
203. Yahoo Finance. Yahoo Finance. Online financial data platform. Accessed: 2025-09-10 (2025).
204. Investing.com. Investing.com: Financial markets portal. Online financial data platform. Accessed: 2025-09-10 (2025).
205. Aroussi, R. yfinance: Open-source Python interface for Yahoo Finance. Open-source software. Accessed: 2025-09-10 (2025).

206. Alpha Vantage. Alpha Vantage: Apis for stock, forex, and cryptocurrency data. Online financial data API. Accessed: 2025-09-10 (2025).
207. Alpaca. Alpaca: Api-first brokerage platform. Online brokerage and market-data API. Accessed: 2025-09-10 (2025).
208. Polygon.io. Polygon.io: Real-time and historical market data APIs. Online market data API. Accessed: 2025-09-10 (2025).
209. Binance. Binance: Cryptocurrency exchange. Online cryptocurrency exchange and data API. Accessed: 2025-09-10 (2025).
210. CoinMarketCap. CoinMarketCap: Cryptocurrency market data aggregator. Online cryptocurrency data platform. Accessed: 2025-09-10 (2025).
211. Aksu, T., Liu, J., Liu, X. & Sahoo, D. GIFT-Eval: A benchmark for general time series forecasting model evaluation. CoRR preprint (2024).
212. Qiu, X., Zhou, L., Wu, X. & Yang, B. TFB: Towards comprehensive and fair benchmarking of time series forecasting methods. *Proc. VLDB Endow*. (2024).
213. López de Prado, M. Advances in Financial Machine Learning (Wiley, 2018).
214. Ramelli, S. & Wagner, A. F. Feverish stock price reactions to COVID-19. *Rev. Corp. Financ. Stud.* **9**, 622–655, https://doi.org/10.1093/rcfs/cfaa012 (2020).
215. Peters, J., Bühlmann, P. & Meinshausen, N. Causal inference by using invariant prediction: identification and confidence intervals. *J. R. Stat. Soc. Ser. B Stat. Methodol.* **78**, 947–1012 (2016).
216. Raissi, M., Perdikaris, P. & Karniadakis, G. E. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *J. Comput. Phys.* **378**, 686–707 (2019).
217. Merton, R. C. Option pricing when underlying stock returns are discontinuous. *J. Financ. Econ.* **3**, 125–144 (1976).
218. Jin, M., Ma, L., Chu, Z. & Wen, Q. Time-LLM: Time series forecasting by reprogramming large language models. In *ICLR* (2024).
219. Nguyen, C. V., Li, Y., Bui, T. D. & Turner, R. E. Variational continual learning. In *International Conference on Learning Representations (ICLR)* (2018).
220. Moody, J. & Saffell, M. Learning to trade via direct reinforcement. *IEEE Trans. Neural Netw.* **12**, 875–889 (2001).
221. Aksu, T. et al. GIFT-eval: A benchmark for general time series forecasting model evaluation. *CoRR* (2024). 2410.10393.
222. Xu, W. et al. FinMultiTime: A four-modal bilingual dataset for financial time-series analysis. *CoRR*. 2506.05019 (2025).

Acknowledgements

The authors would like to thank the colleagues and members of the research group for constructive discussions that contributed to refining the scope and taxonomy of this survey.

Author contributions

Y. Zhang: Conceptualization, Methodology, Writing -- Original Draft. J. Duan: Validation, Formal analysis, Writing -- Review & Editing. Y. Zou: Validation, Formal analysis, Writing -- Review & Editing. K. Feng: Supervision, Project administration. J. Xiao: Supervision, Project administration. All authors reviewed, discussed, and approved the final submitted version of the manuscript.

Funding

The authors declare that no funds, grants, or other financial support were received during the preparation of this manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s44387-026-00166-3.

Correspondence and requests for materials should be addressed to Ke Feng or Jing Xiao.

Reprints and permissions information is available at http://www.nature.com/reprints

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026

This Article in Press is shared early to give you faster access to new research. It is citable and carries a permanent DOI. The final edited version will replace it automatically.